# DRV425 Fluxgate Magnetic-Field Sensor

## 1 Features

- High-precision, integrated fluxgate sensor:
  - Offset: ±8µT (max)
  - Offset drift: ±5nT/°C (typ)
  - Gain error: 0.04% (typ)
  - Gain drift: ±7ppm/°C (typ)
  - Linearity: ±0.1%
  - Noise: 1.5nT/$\sqrt{\text{Hz}}$ (typ)
- Sensor range: ±2mT (max)
  - Range and gain adjustable with external resistor
- Selectable bandwidth: 47kHz or 32kHz
- Precision reference:
  - Accuracy: 2% (max), drift: 50ppm/°C (max)
  - Pin-selectable voltage: 2.5V or 1.65V
  - Selectable ratiometric mode: VDD / 2
- Diagnostic features: Overrange and error flags
- Supply voltage range: 3.0V to 5.5V

## 2 Applications

- Linear position sensing
- Current sensing in busbars
- Over-the-trace current sensing
- General-purpose magnetic-field sensors
- Overcurrent detection
- Motor reliability diagnostics
- Frequency and voltage inverters
- Solar inverters

## 3 Description

The DRV425 is designed for single-axis magnetic field-sensing applications, and enables electrically-isolated, high-sensitivity, and precise dc- and ac-field measurements. The device provides the unique and proprietary, integrated fluxgate sensor (IFG) with an internal compensation coil to support a high-accuracy sensing range of ±2mT, with a measurement bandwidth of up to 47kHz. The low offset, offset drift, and noise of the sensor, combined with the precise gain, low gain drift, and very low nonlinearity provided by the internal compensation coil, result in unrivaled magnetic field measurement precision. The output of the DRV425 is an analog signal proportional to the sensed magnetic field.

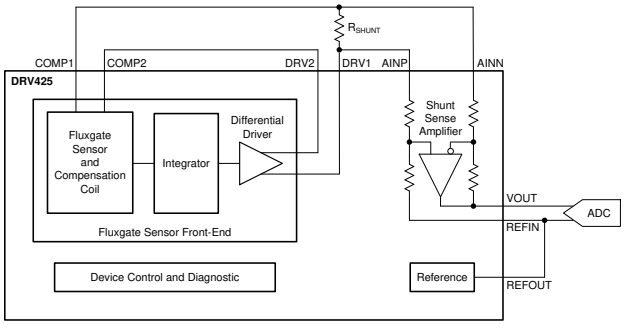
The device offers a complete set of features, including an internal difference amplifier, on-chip precision reference, and diagnostic functions to minimize component count and system-level cost.

The device is available in a thermally-enhanced, nonmagnetic, thin WQFN package, with a thermal pad for optimized heat dissipation, and is specified for operation over the industrial temperature range of –40°C to +125°C.

**Package Information**

| PART NUMBER | PACKAGE(1) | PACKAGE SIZE(2) |
|---|---|---|
| DRV425 | RTJ (WQFN, 20) | 4.00mm × 4.00mm |

(1) For more information, see Section 10.
(2) The package size (length × width) is a nominal value and includes pins, where applicable.




**Simplified Schematic**

An IMPORTANT NOTICE at the end of this data sheet addresses availability, warranty, changes, use in safety-critical applications, intellectual property matters and other important disclaimers. PRODUCTION DATA.

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# 4 Pin Configuration and Functions

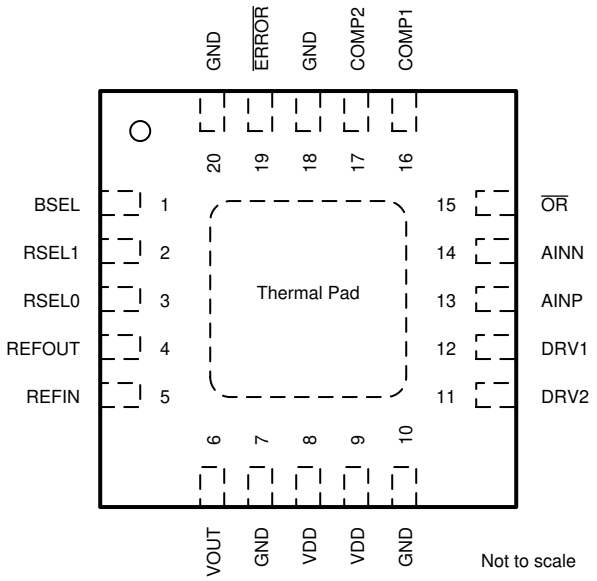



**Figure 4-1. RTJ Package 20-Pin WQFN Top View**

**Table 4-1. Pin Functions**

| PIN | | Type[1] | DESCRIPTION |
|---|---|---|---|
| NAME | NO. | | |
| AINN | 14 | I | Inverting input of the shunt-sense amplifier |
| AINP | 13 | I | Noninverting input of the shunt-sense amplifier |
| BSEL | 1 | I | Filter bandwidth select input |
| COMP1 | 16 | I | Internal compensation coil input 1 |
| COMP2 | 17 | I | Internal compensation coil input 2 |
| DRV1 | 12 | O | Compensation coil driver output 1 |
| DRV2 | 11 | O | Compensation coil driver output 2 |
| $\overline{\text{ERROR}}$ | 19 | O | Error flag: open-drain, active-low output |
| GND | 7, 10, 18, 20 | — | Ground reference |
| $\overline{\text{OR}}$ | 15 | O | Shunt-sense amplifier overrange indicator: open-drain, active-low output |
| REFIN | 5 | I | Common-mode reference input for the shunt-sense amplifier |
| REFOUT | 4 | O | Voltage reference output |
| RSEL0 | 3 | I | Voltage reference mode selection input 0 |
| RSEL1 | 2 | I | Voltage reference mode selection input 1 |
| VDD | 8, 9 | — | Supply voltage, 3.0V to 5.5V. Decouple both pins using 1µF ceramic capacitors placed as close as possible to the device. See the *Power Supply Decoupling* and *Layout* sections for further details. |
| VOUT | 6 | O | Shunt-sense amplifier output |
| Thermal Pad | Thermal Pad | — | Connect the thermal pad to GND |

(1) I = Input; O = Output

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# 5 Specifications

## 5.1 Absolute Maximum Ratings

over operating free-air temperature range (unless otherwise noted)(1)

| | | MIN | MAX | UNIT |
|---|---|---|---|---|
| Voltage | Supply voltage (VDD to GND) | –0.3 | 6.5 | V |
| | Input voltage, except AINP and AINN pins(2) | GND – 0.5 | VDD + 0.5 | |
| | Shunt-sense amplifier inputs (AINP and AINN pins)(3) | GND – 6.0 | VDD + 6.0 | |
| Current | DRV1 and DRV2 pins (short-circuit current, $I_{OS}$)(4) | –300 | 300 | mA |
| | Shunt-sense amplifier input pins AINP and AINN | –5 | 5 | |
| | All remaining pins | –25 | 25 | |
| Temperature | Junction, $T_J$ | –50 | 150 | °C |
| | Storage, $T_{stg}$ | –65 | 150 | |

(1) Operation outside the *Absolute Maximum Ratings* may cause permanent device damage. *Absolute Maximum Ratings* do not imply functional operation of the device at these or any other conditions beyond those listed under *Recommended Operating Conditions*. If used outside the *Recommended Operating Conditions* but within the *Absolute Maximum Ratings*, the device may not be fully functional, and this may affect device reliability, functionality, performance, and shorten the device lifetime.
(2) Input pins are diode-clamped to the power-supply rails. Input signals that can swing more than 0.5V beyond the supply rails must be current limited, except for the differential amplifier input pins.
(3) These inputs are not diode-clamped to the power-supply rails.
(4) Power-limited; observe maximum junction temperature.

## 5.2 ESD Ratings

| | | | VALUE | UNIT |
|---|---|---|---|---|
| $V_{(ESD)}$ | Electrostatic discharge | Human-body model (HBM), per ANSI/ESDA/JEDEC JS-001(1)<br>HBM ESD classification level 2 | ±2000 | V |
| | | Charged-device model (CDM), per JEDEC specification JESD22C101(2)<br>CDM ESD classification level C6 | ±1000 | |

(1) JEDEC document JEP155 states that 500V HBM allows safe manufacturing with a standard ESD control process.
(2) JEDEC document JEP157 states that 250V CDM allows safe manufacturing with a standard ESD control process.

## 5.3 Recommended Operating Conditions

over operating free-air temperature range (unless otherwise noted)

| | | MIN | NOM | MAX | UNIT |
|---|---|---|---|---|---|
| VDD | Supply voltage range (VDD to GND) | 3.0 | 5.0 | 5.5 | V |
| $T_A$ | Specified ambient temperature | –40 | | 125 | °C |

## 5.4 Thermal Information

| THERMAL METRIC(1) | | DRV425<br>RTJ (WQFN)<br>20 PINS | UNIT |
|---|---|---|---|
| $R_{\theta JA}$ | Junction-to-ambient thermal resistance | 34.1 | °C/W |
| $R_{\theta JC(top)}$ | Junction-to-case (top) thermal resistance | 33.1 | °C/W |
| $R_{\theta JB}$ | Junction-to-board thermal resistance | 11 | °C/W |
| $\psi_{JT}$ | Junction-to-top characterization parameter | 0.3 | °C/W |
| $\psi_{JB}$ | Junction-to-board characterization parameter | 11 | °C/W |
| $R_{\theta JC(bot)}$ | Junction-to-case (bottom) thermal resistance | 2.1 | °C/W |

(1) For more information about traditional and new thermal metrics, see the *Semiconductor and IC Package Thermal Metrics* application note.


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## 5.5 Electrical Characteristics

all minimum and maximum specifications are at $T_A$ = 25°C, VDD = 3.0V to 5.5V, and $I_{DRV1}$ = $I_{DRV2}$ = 0mA (unless otherwise noted); typical values are at VDD = 5.0V.

| PARAMETER | | | TEST CONDITIONS | MIN | TYP | MAX | UNIT |
|---|---|---|---|---|---|---|---|
| **FLUXGATE SENSOR FRONT-END** | | | | | | | |
| | Offset | | No magnetic field | –8 | ±2 | 8 | µT |
| | Offset drift | | No magnetic field | | ±5 | | nT/°C |
| G | Gain | | Current at DRV1 and DRV2 outputs | | 12.2 | | mA/mT |
| | Gain error | | | | ±0.04% | | |
| | Gain drift | | Best-fit line method | | ±7 | | ppm/°C |
| | Linearity error | | | | 0.1% | | |
| | Hysteresis | | Magnetic field sweep from –10mT to 10mT | | 1.4 | | µT |
| | Noise | | f = 0.1Hz to 10Hz | | 17 | | nTrms |
| | Noise density | | f = 1kHz | | 1.5 | | nT/√Hz |
| | Compensation range | | | –2 | | 2 | mT |
| | Saturation trip level for the ERROR pin[2] | | Open-loop, uncompensated field | | 1.6 | | mT |
| | ERROR delay | | Open-loop at B > 1.6mT | | 4 to 6 | | µs |
| BW | Bandwidth | | BSEL = 0, $R_{SHUNT}$ = 22Ω | | 32 | | kHz |
| | | | BSEL = 1, $R_{SHUNT}$ = 22Ω | | 47 | | |
| $I_{OS}$ | Short-circuit current | | VDD = 5V | | 250 | | mA |
| | | | VDD = 3.3V | | 150 | | |
| | Common-mode output voltage at the DRV1 and DRV2 pins | | | | $V_{REFOUT}$ | | V |
| | Compensation coil resistance | | | | 100 | | Ω |
| **SHUNT-SENSE AMPLIFIER** | | | | | | | |
| $V_{OO}$ | Output offset voltage | | $V_{AINP}$ = $V_{AINN}$ = $V_{REFIN}$, VDD = 3.0V | –0.075 | ±0.01 | 0.075 | mV |
| | Output offset voltage drift | | | –2 | ±0.4 | 2 | µV/°C |
| CMRR | Common-mode rejection ratio, RTO[1] | | $V_{CM}$ = –1V to VDD + 1V, $V_{REFIN}$ = VDD / 2 | –250 | ±50 | 250 | µV/V |
| $PSRR_{AMP}$ | Power-supply rejection ratio, RTO[1] | | VDD = 3.0V to 5.5V, $V_{CM}$ = $V_{REFIN}$ | –86 | ±4 | 86 | µV/V |
| $V_{ICR}$ | Common-mode input voltage range | | | –1 | | VDD + 1 | V |
| $z_{id}$ | Differential input impedance | | | 16.5 | 20 | 23.5 | kΩ |
| $z_{ic}$ | Common-mode input impedance | | | 40 | 50 | 60 | kΩ |
| $G_{nom}$ | Nominal gain | | $V_{VOUT}$ / ($V_{AINP}$ – $V_{AINN}$) | | 4 | | V/V |
| $E_G$ | Gain error | | | –0.3% | ±0.02% | 0.3% | |
| | Gain error drift | | | –5 | ±1 | 5 | ppm/°C |
| | Linearity error | | | | 12 | | ppm |
| | Voltage output swing from negative rail (OR pin trip level)[2] | | VDD = 5.5V, $I_{VOUT}$ = 2.5mA | | 48 | 85 | mV |
| | | | VDD = 3.0V, $I_{VOUT}$ = 2.5mA | | 56 | 100 | |
| | Voltage output swing from positive rail (OR pin trip level)[2] | | VDD = 5.5V, $I_{VOUT}$ = –2.5mA | VDD – 85 | VDD – 48 | | mV |
| | | | VDD = 3.0V, $I_{VOUT}$ = –2.5mA | VDD – 100 | VDD – 56 | | |
| | Signal overrange indication delay (OR pin)[2] | | $V_{IN}$ = 1V step | | 2.5 to 3.5 | | µs |
| $I_{OS}$ | Short-circuit current | | VOUT connected to GND | | –18 | | mA |
| | | | VOUT connected to VDD | | 20 | | |
| $BW_{-3dB}$ | Bandwidth | | | | 2 | | MHz |
| SR | Slew rate | | | | 6.5 | | V/µs |
| $t_{sa}$ | Settling time | Large signal | ΔV = ± 2V to 1%, no external filter | | 0.9 | | µs |
| | | Small signal | ΔV = ± 0.4V to 0.01% | | 8 | | |
| $e_n$ | Output voltage noise density | | f = 1kHz, compensation loop disabled | | 170 | | nV/√Hz |
| $V_{REFIN}$ | Input voltage range at pin REFIN | | Input voltage range at REFIN pin | GND | | VDD | V |

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all minimum and maximum specifications are at $T_A$ = 25°C, VDD = 3.0V to 5.5V, and $I_{DRV1}$ = $I_{DRV2}$ = 0mA (unless otherwise noted); typical values are at VDD = 5.0V.

| PARAMETER | | TEST CONDITIONS | MIN | TYP | MAX | UNIT |
|---|---|---|---|---|---|---|
| **VOLTAGE REFERENCE** | | | | | | |
| $V_{REFOUT}$ | Reference output voltage at the REFOUT pin | RSEL[1:0] = 00, no load | 2.45 | 2.5 | 2.55 | V |
| | | RSEL[1:0] = 01, no load | 1.6 | 1.65 | 1.7 | |
| | | RSEL[1:0] = 1x, no load | 45 | 50 | 55 | % of VDD |
| | Reference output voltage drift | RSEL[1:0] = 0x | –50 | ±10 | 50 | ppm/°C |
| | Voltage divider gain error drift | RSEL[1:0] = 1x | –50 | ±10 | 50 | ppm/°C |
| $PSRR_{REF}$ | Power-supply rejection ratio | RSEL[1:0] = 0x | –300 | ±15 | 300 | µV/V |
| $\Delta V_{O(\Delta IO)}$ | Load regulation | RSEL[1:0] = 0x, load to GND or VDD, $\Delta I_{LOAD}$ = 0mA to 5mA, $T_A$ = –40°C to +125°C | | 0.15 | 0.35 | mV/mA |
| | | RSEL[1:0] = 1x, load to GND or VDD, $\Delta I_{LOAD}$ = 0mA to 5mA, $T_A$ = –40°C to +125°C | | 0.3 | 0.8 | |
| $I_{OS}$ | Short-circuit current | REFOUT connected to VDD | | 20 | | mA |
| | | REFOUT connected to GND | | –18 | | mA |
| **DIGITAL INPUTS/OUTPUTS (CMOS)** | | | | | | |
| $I_{IL}$ | Input leakage current | | | 0.01 | | µA |
| $V_{IH}$ | High-level input voltage | $T_A$ = –40°C to +125°C | 0.7 × VDD | | VDD + 0.3 | V |
| $V_{IL}$ | Low-level input voltage | $T_A$ = –40°C to +125°C | –0.3 | | 0.3 × VDD | V |
| $V_{OH}$ | High-level output voltage | Open-drain output | Set by external pullup resistor | | | V |
| $V_{OL}$ | Low-level output voltage | 4mA sink current | | 0.3 | | V |
| **POWER SUPPLY** | | | | | | |
| $I_Q$ | Quiescent current | $I_{DRV1/2}$ = 0mA, 3.0V ≤ VDD ≤ 3.6V, $T_A$ = –40°C to +125°C | | 6 | 8 | mA |
| | | $I_{DRV1/2}$ = 0mA, 4.5V ≤ VDD ≤ 5.5V, $T_A$ = –40°C to +125°C | | 7 | 10 | |
| $V_{POR}$ | Power-on reset threshold | | | 2.4 | | V |

(1) Parameter value is referred-to-output (RTO).
(2) See the *Magnetic Field Range, Overrange Indicator, and Error Flag* section for details on the behavior of the ERROR and OR outputs.


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## 5.6 Typical Characteristics

at VDD = 5V and $T_A$ = 25°C (unless otherwise noted)

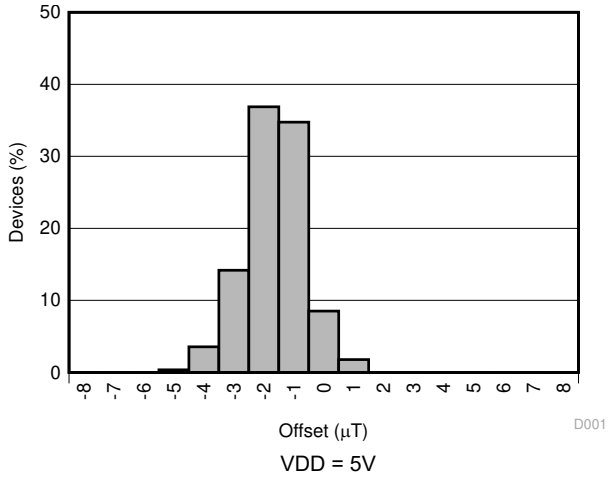



**Figure 5-1. Fluxgate Sensor Front-end Offset Histogram**

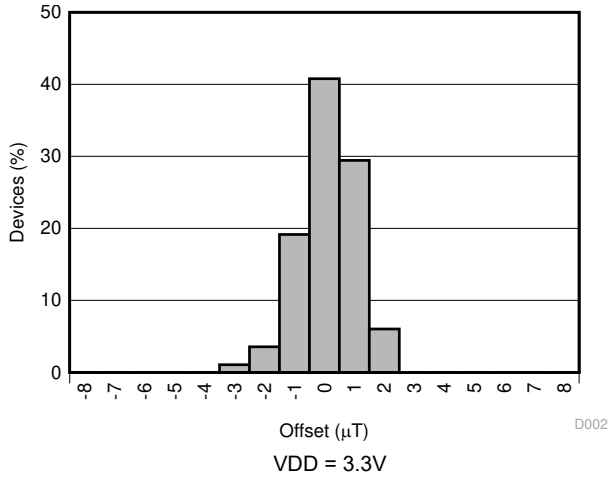



**Figure 5-2. Fluxgate Sensor Front-end Offset Histogram**

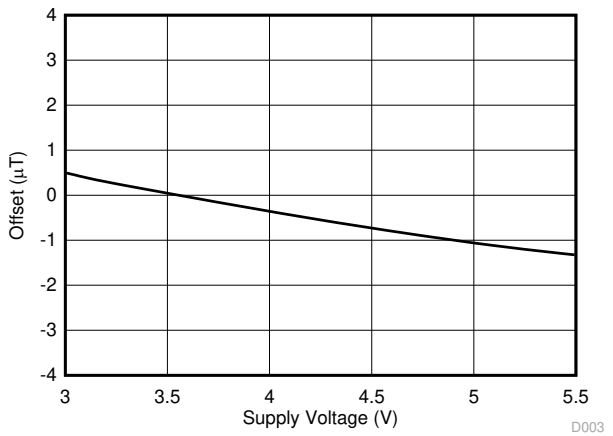



**Figure 5-3. Fluxgate Sensor Front-end Offset vs Supply Voltage**

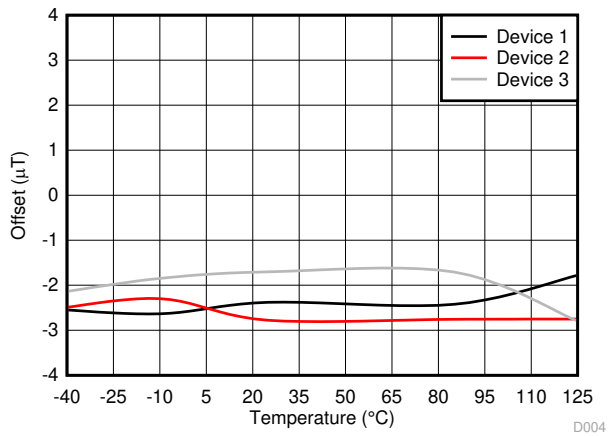



**Figure 5-4. Fluxgate Sensor Front-end Offset vs Temperature**

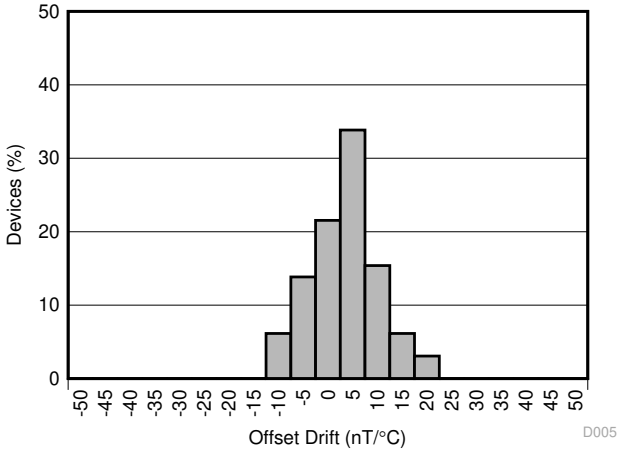



**Figure 5-5. Fluxgate Sensor Front-end Offset Drift Histogram**

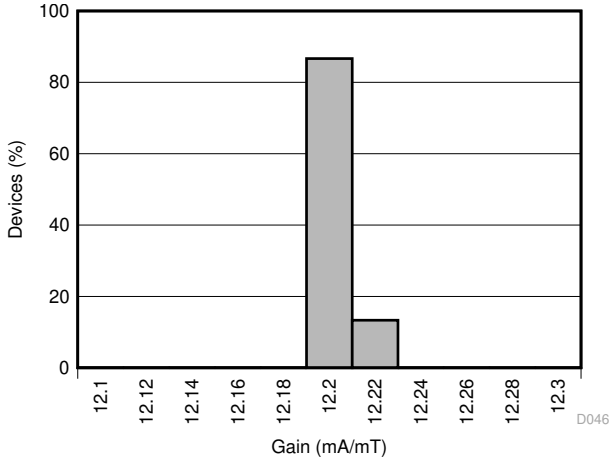



**Figure 5-6. Fluxgate Sensor Front-end Gain Histogram**

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## 5.6 Typical Characteristics (continued)

at VDD = 5V and $T_A$ = 25°C (unless otherwise noted)

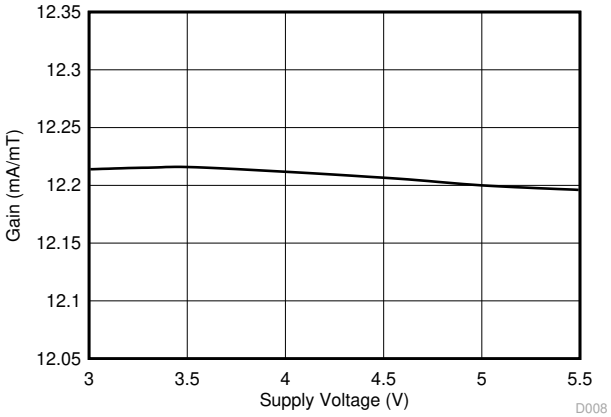



**Figure 5-7. Fluxgate Sensor Front-end Gain vs Supply Voltage**

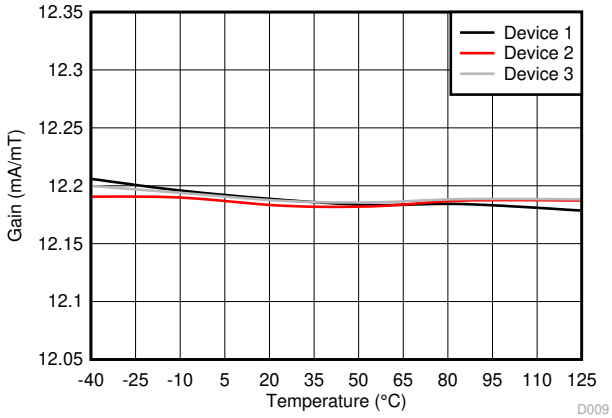



**Figure 5-8. Fluxgate Sensor Front-end Gain vs Temperature**

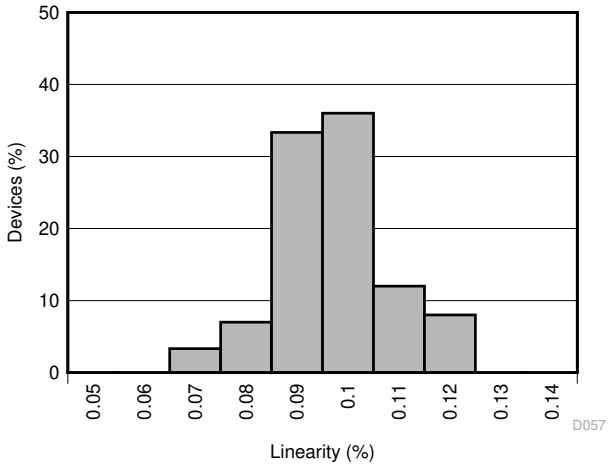



**Figure 5-9. Fluxgate Sensor Front-end Linearity Histogram**

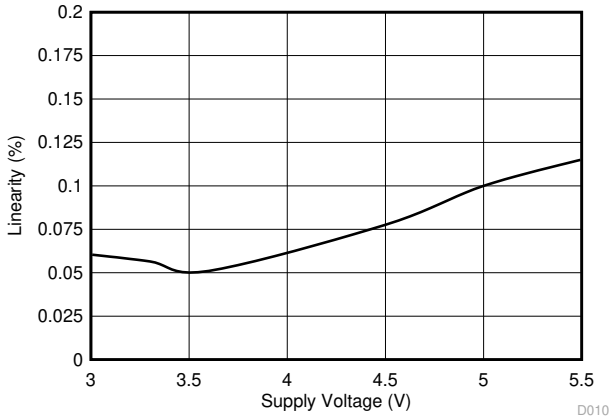



**Figure 5-10. Fluxgate Sensor Front-end Linearity vs Supply Voltage**

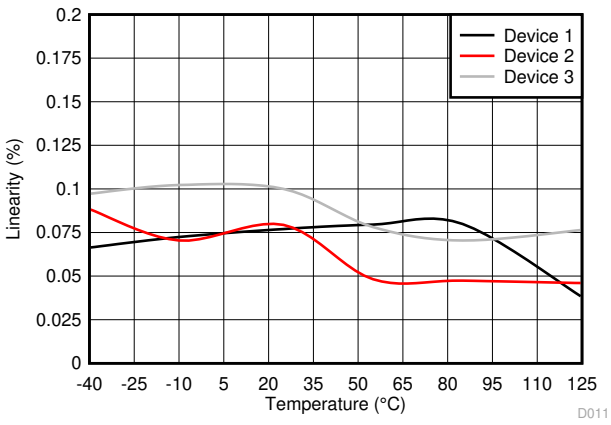



**Figure 5-11. Fluxgate Sensor Front-end Linearity vs Temperature**

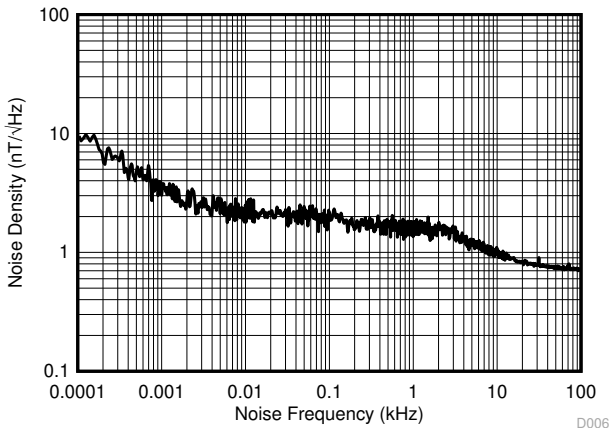



**Figure 5-12. Fluxgate Sensor Front-end Noise Density vs Noise Frequency**


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## 5.6 Typical Characteristics (continued)

at VDD = 5V and $T_A$ = 25°C (unless otherwise noted)

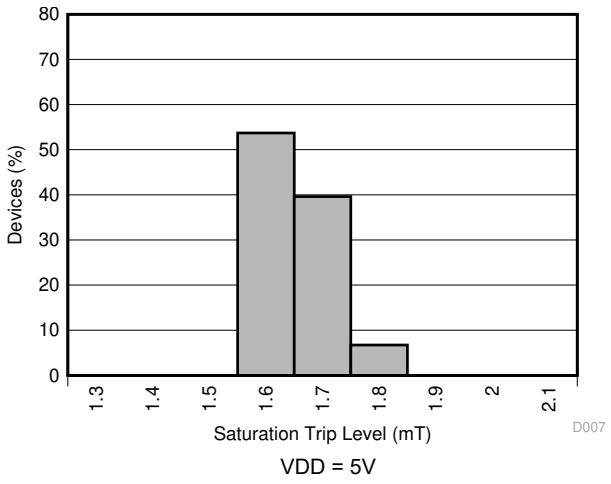



**Figure 5-13. Fluxgate Sensor Saturation ( $\overline{\text{ERROR}}$ Pin) Trip Level Histogram**

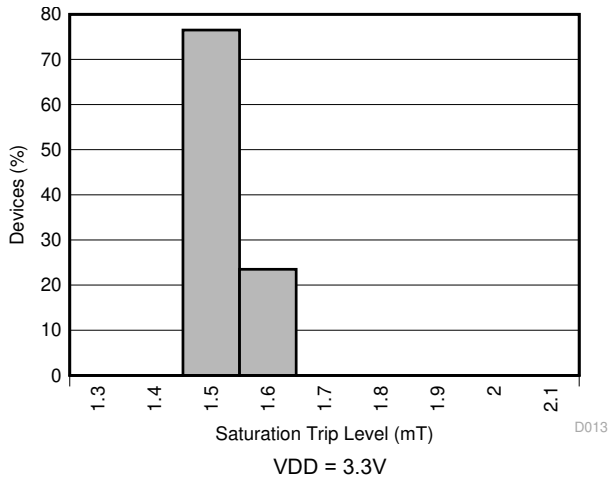



**Figure 5-14. Fluxgate Sensor Saturation ( $\overline{\text{ERROR}}$ Pin) Trip Level Histogram**

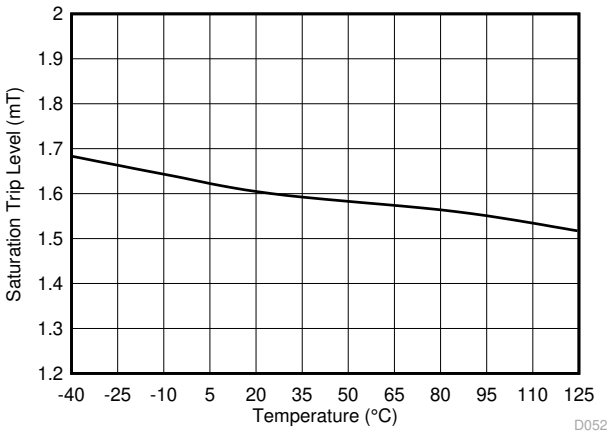



**Figure 5-15. Fluxgate Sensor Saturation ( $\overline{\text{ERROR}}$ Pin) Trip Level vs Temperature**

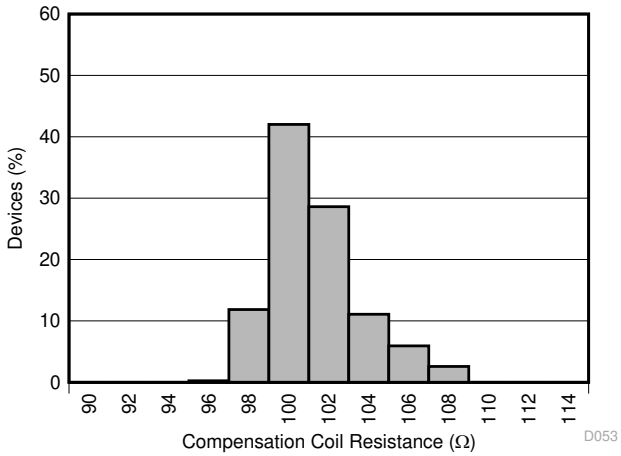



**Figure 5-16. Compensation Coil Resistance Histogram**

## 5.6 Typical Characteristics (continued)

at VDD = 5V and $T_A$ = 25°C (unless otherwise noted)

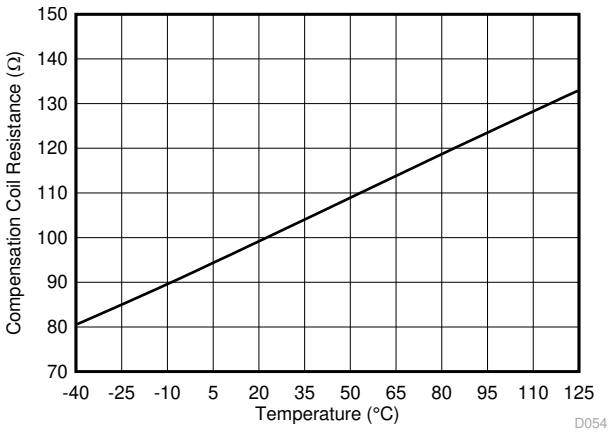



**Figure 5-17. Compensation Coil Resistance vs Temperature**

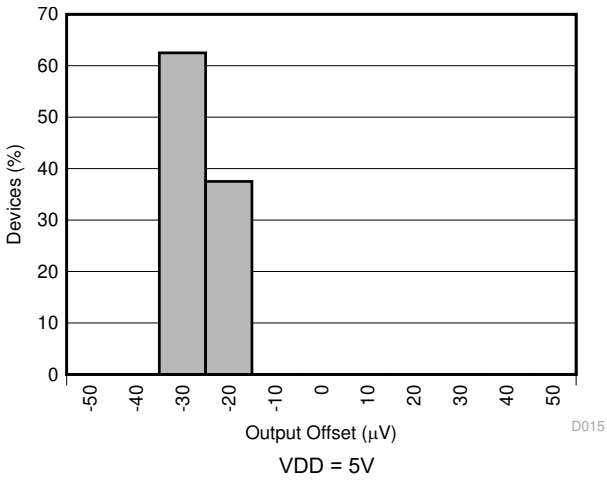



**Figure 5-18. Shunt-Sense Amplifier Output Offset Histogram**

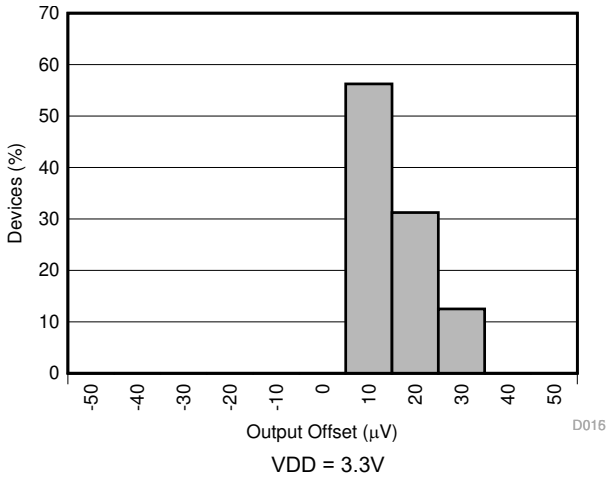



**Figure 5-19. Shunt-Sense Amplifier Output Offset Histogram**

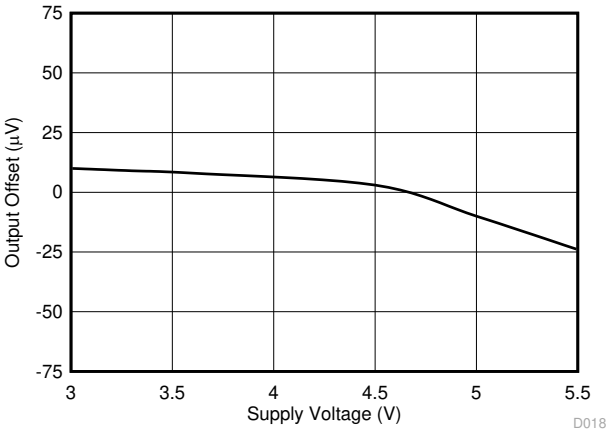



**Figure 5-20. Shunt-Sense Amplifier Output Offset vs Supply Voltage**

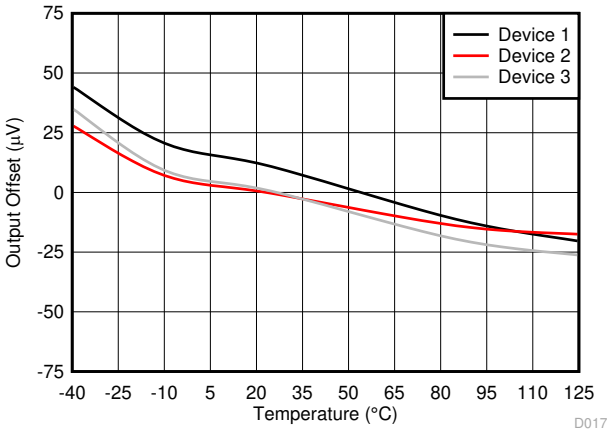



**Figure 5-21. Shunt-Sense Amplifier Output Offset vs Temperature**

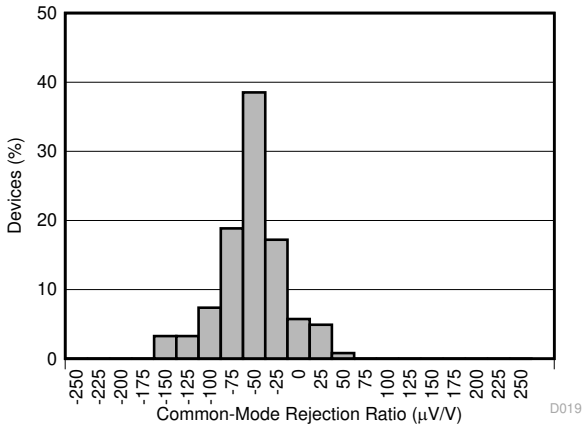



**Figure 5-22. Shunt-Sense Amplifier CMRR Histogram**


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## 5.6 Typical Characteristics (continued)

at VDD = 5V and $T_A$ = 25°C (unless otherwise noted)

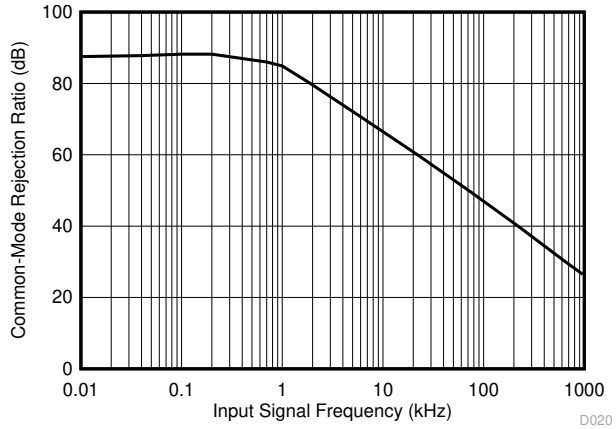



**Figure 5-23. Shunt-Sense Amplifier CMRR vs Input Signal Frequency**

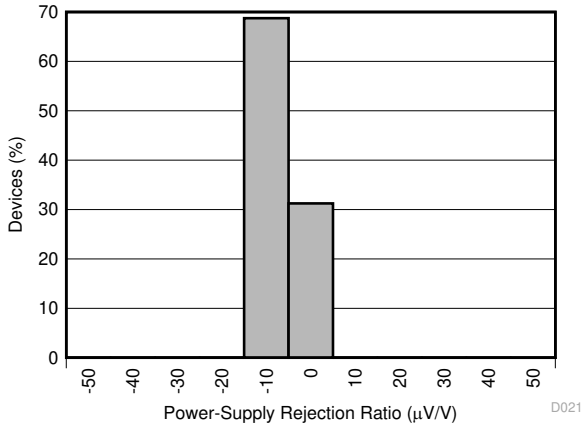



**Figure 5-24. Shunt-Sense Amplifier PSRR Histogram**

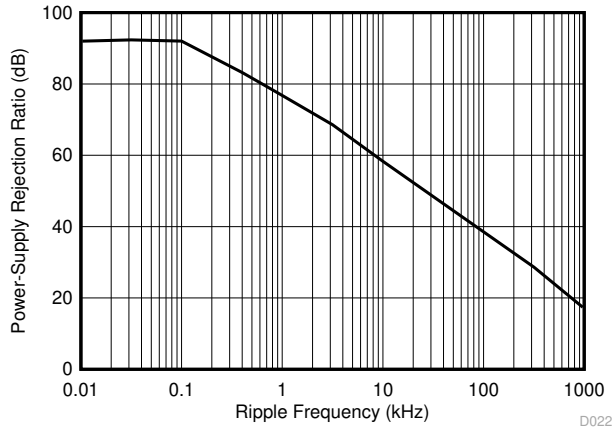



**Figure 5-25. Shunt-Sense Amplifier PSRR vs Ripple Frequency**

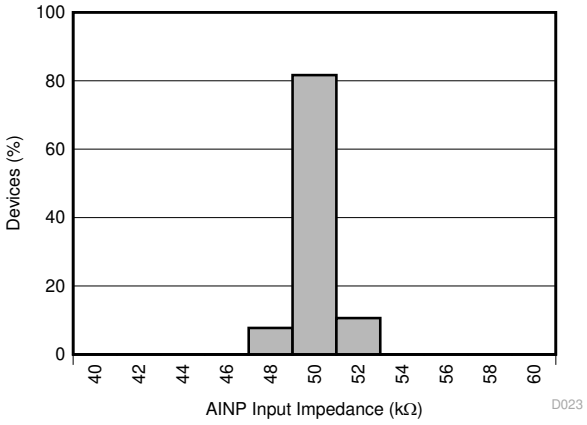



**Figure 5-26. Shunt-Sense Amplifier AINP Input Impedance Histogram**

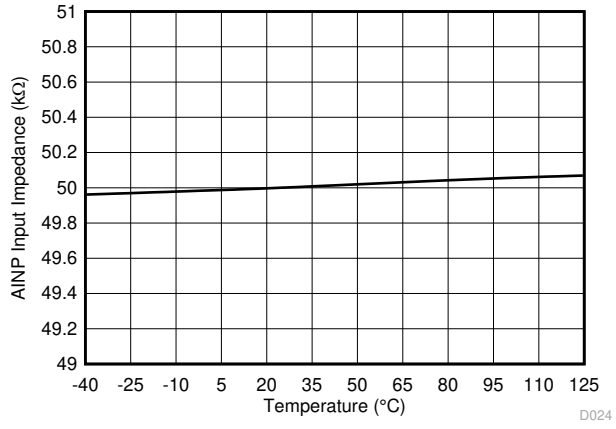



**Figure 5-27. Shunt-Sense Amplifier AINP Input Impedance vs Temperature**

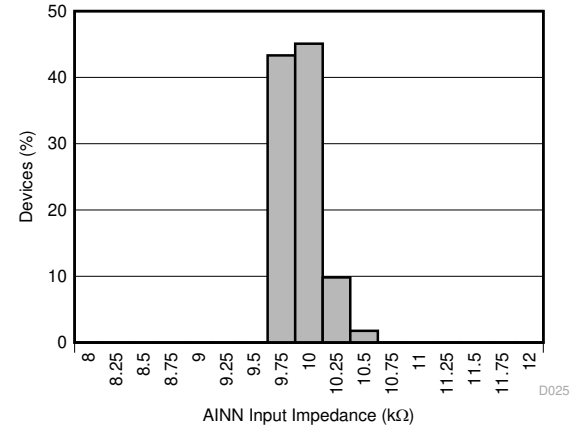



**Figure 5-28. Shunt-Sense Amplifier AINN Input Impedance Histogram**

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## 5.6 Typical Characteristics (continued)

at VDD = 5V and $T_A$ = 25°C (unless otherwise noted)

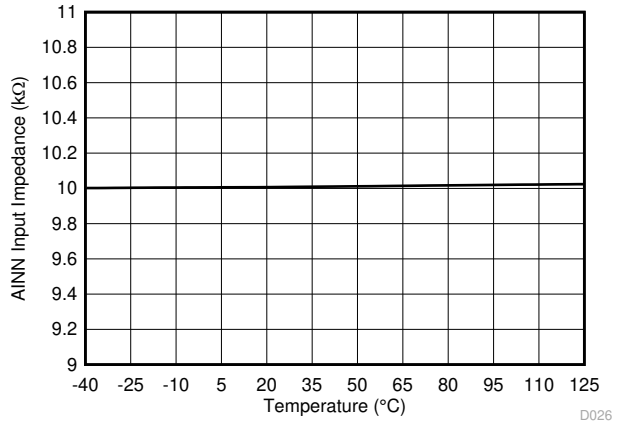
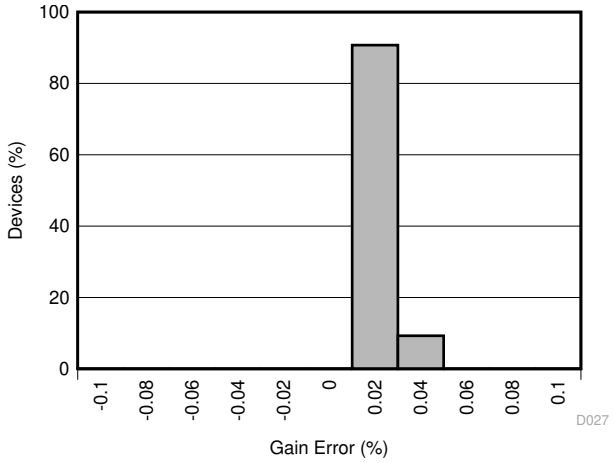
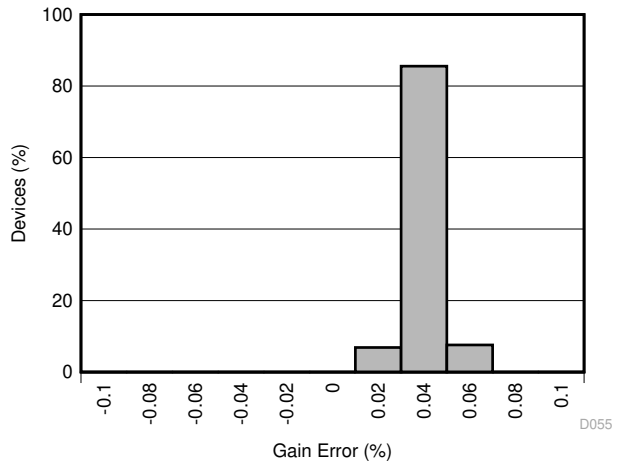



**Figure 5-29. Shunt-Sense Amplifier AINN Input Impedance vs Temperature**




Including IFG, VDD = 5V

**Figure 5-30. Shunt-Sense Amplifier Gain Error Histogram**




Including IFG, VDD = 3.3V

**Figure 5-31. Shunt-Sense Amplifier Gain Error Histogram**

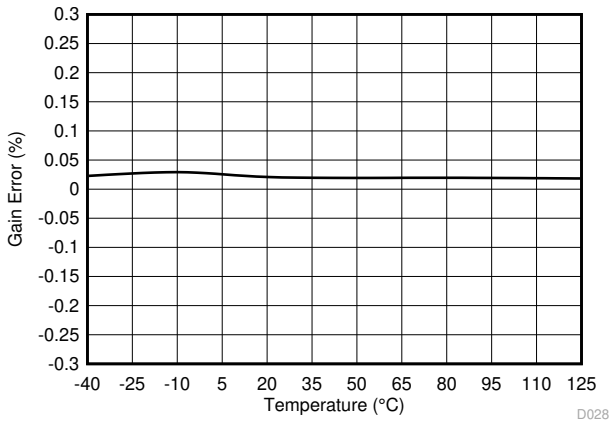



**Figure 5-32. Shunt-Sense Amplifier Gain Error vs Temperature**


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## 5.6 Typical Characteristics (continued)

at VDD = 5V and $T_A$ = 25°C (unless otherwise noted)

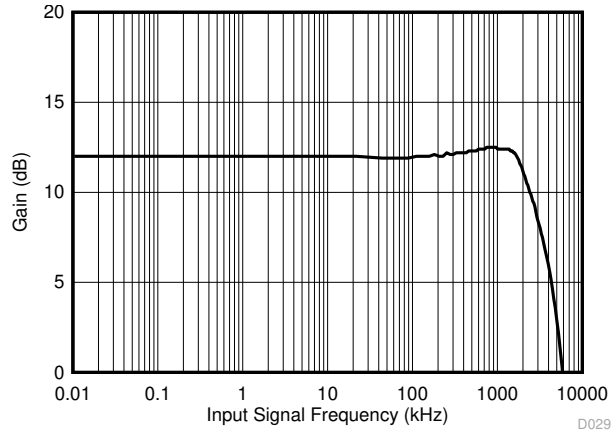



**Figure 5-33. Shunt-Sense Amplifier Gain vs Input Signal Frequency**

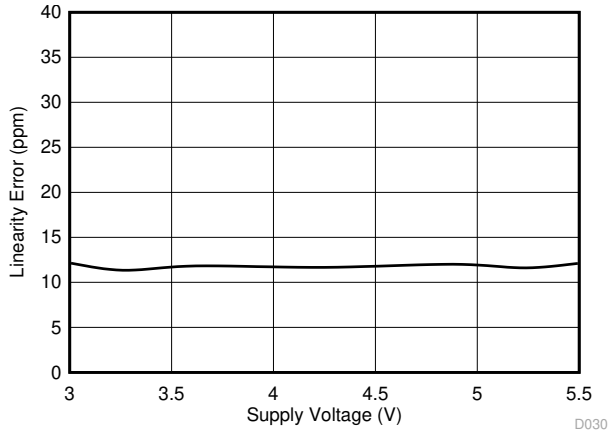



**Figure 5-34. Shunt-Sense Amplifier Linearity Error vs Supply Voltage**

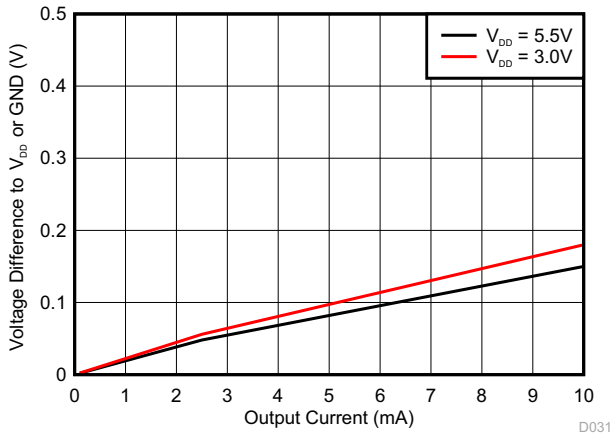



**Figure 5-35. $\overline{OR}$ Pin Trip Level vs Output Current**

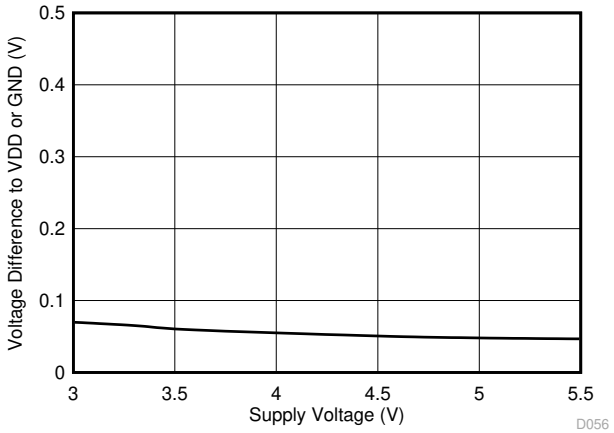



**Figure 5-36. $\overline{OR}$ Pin Trip Level vs Supply Voltage**

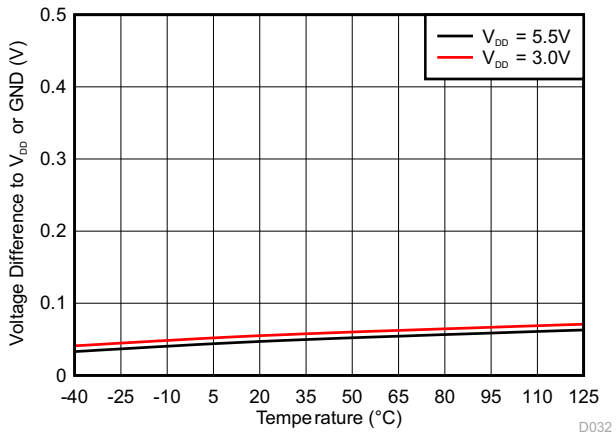



**Figure 5-37. $\overline{OR}$ Pin Trip Level vs Temperature**

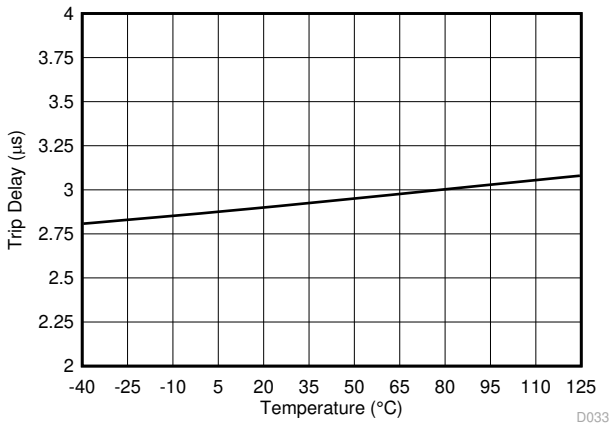



**Figure 5-38. $\overline{OR}$ Pin Trip Delay vs Temperature**

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## 5.6 Typical Characteristics (continued)

at VDD = 5V and $T_A$ = 25°C (unless otherwise noted)

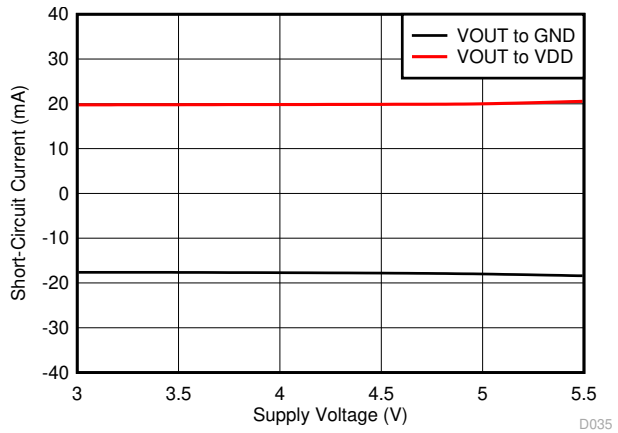



**Figure 5-39. Shunt-Sense Amplifier Output Short-Circuit Current vs Supply Voltage**

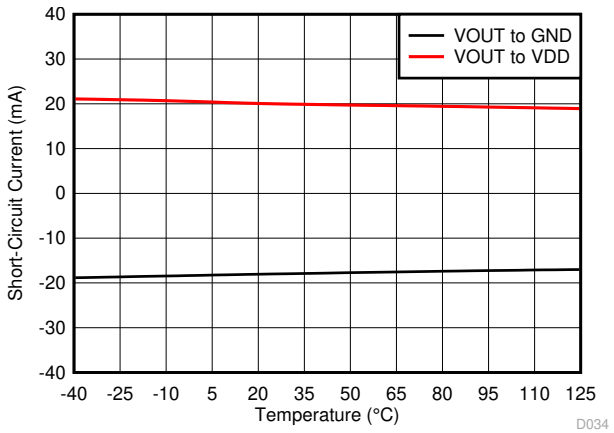



**Figure 5-40. Shunt-Sense Amplifier Output Short-Circuit Current vs Temperature**

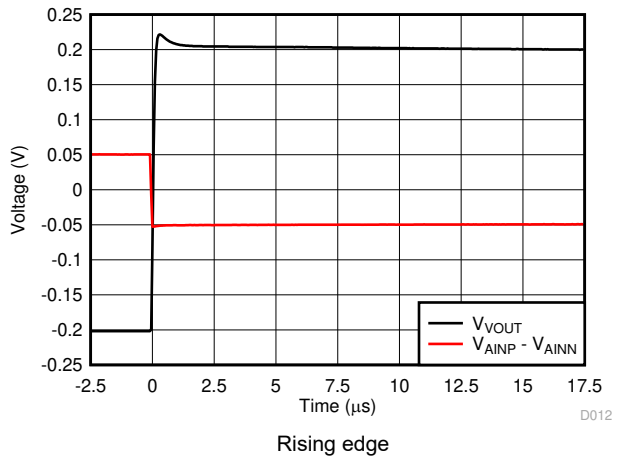



**Figure 5-41. Shunt-Sense Amplifier Small-Signal Settling Time**

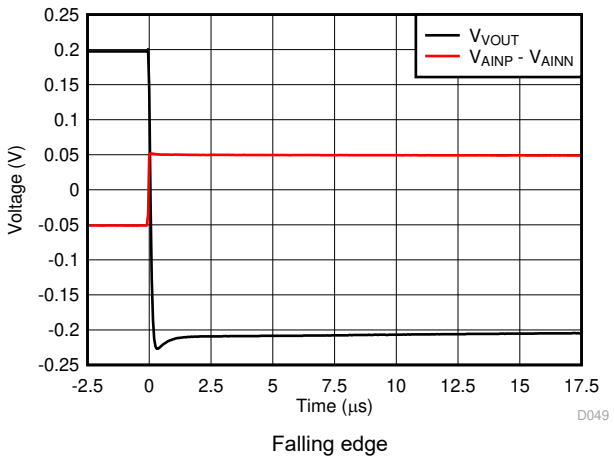



**Figure 5-42. Shunt-Sense Amplifier Small-Signal Settling Time**

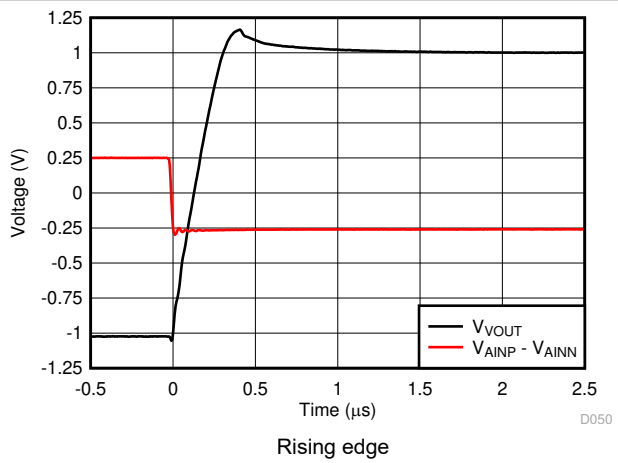



**Figure 5-43. Shunt-Sense Amplifier Large-Signal Settling Time**

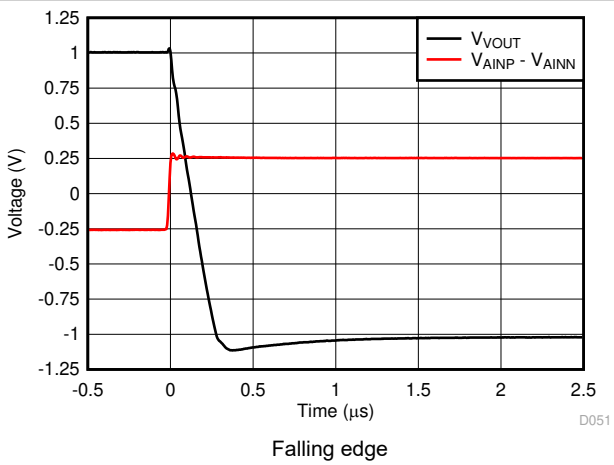
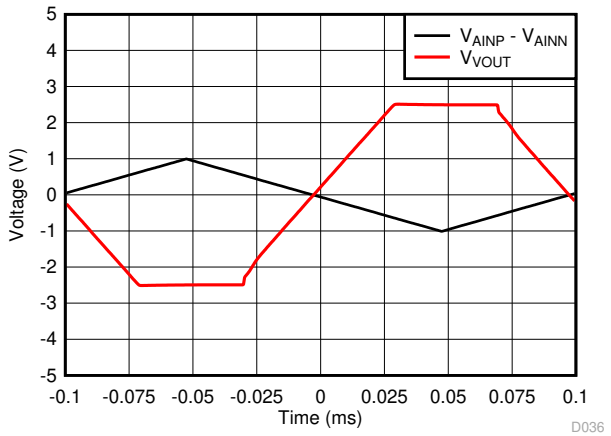



**Figure 5-44. Shunt-Sense Amplifier Large-Signal Settling Time**


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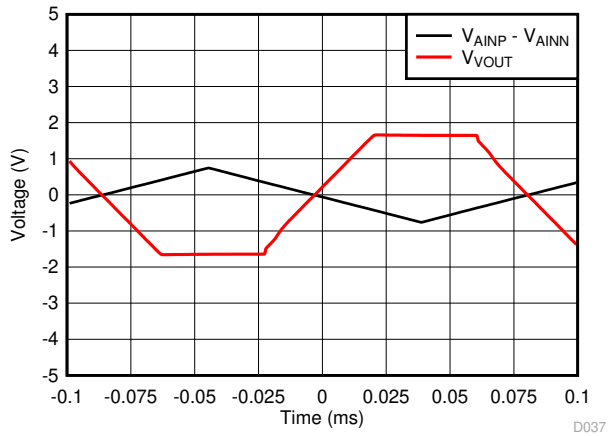
## 5.6 Typical Characteristics (continued)

at VDD = 5V and $T_A$ = 25°C (unless otherwise noted)




VDD = 5V

**Figure 5-45. Shunt-Sense Amplifier Overload Recovery Response**




VDD = 3.3V

**Figure 5-46. Shunt-Sense Amplifier Overload Recovery Response**

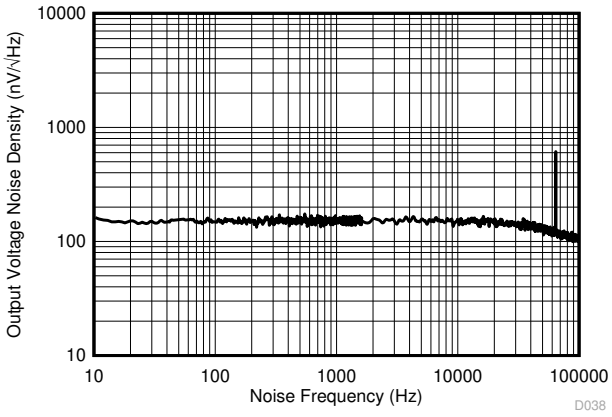
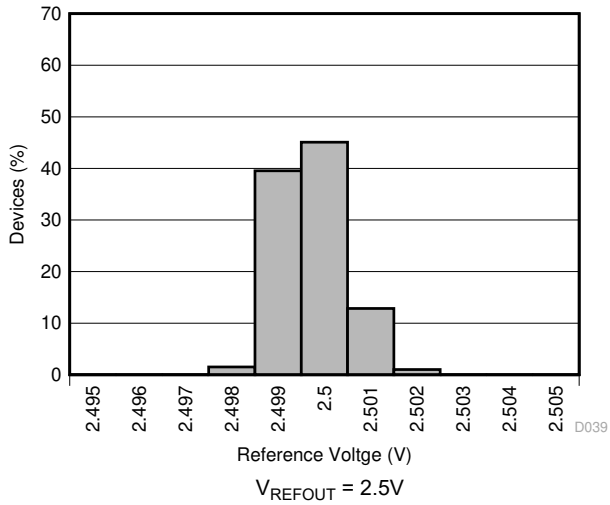



**Figure 5-47. Shunt-Sense Amplifier Output Voltage Noise Density vs Noise Frequency**




$V_{REFOUT}$ = 2.5V

**Figure 5-48. Reference Voltage Histogram**

## 5.6 Typical Characteristics (continued)

at VDD = 5V and $T_A$ = 25°C (unless otherwise noted)

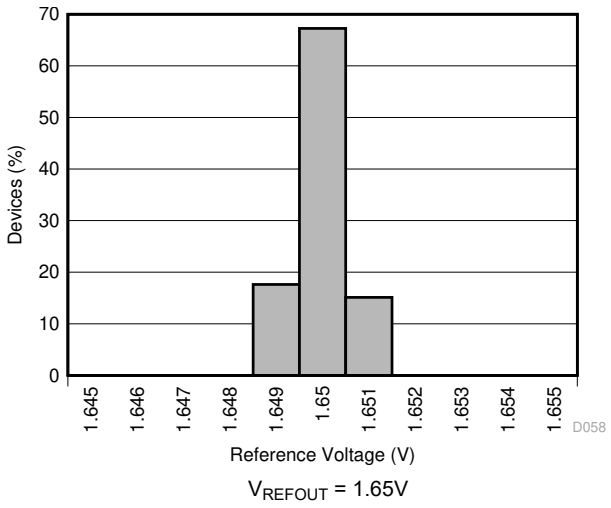



**Figure 5-49. Reference Voltage Histogram**

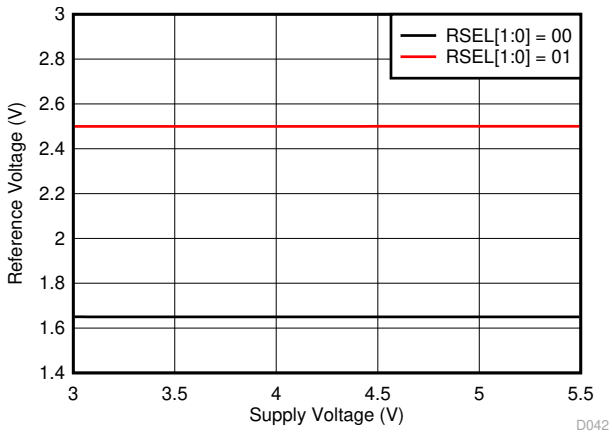



**Figure 5-50. Reference Voltage vs Supply Voltage**

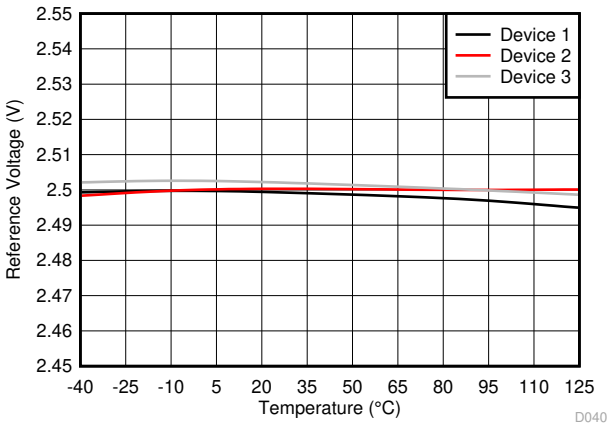



**Figure 5-51. Reference Voltage vs Temperature**

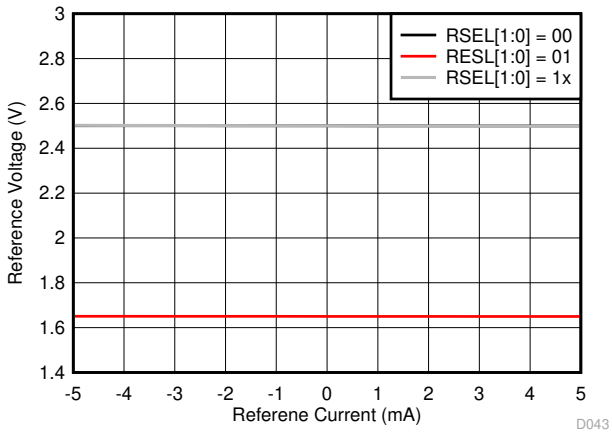



**Figure 5-52. Reference Voltage vs Reference Output Current**

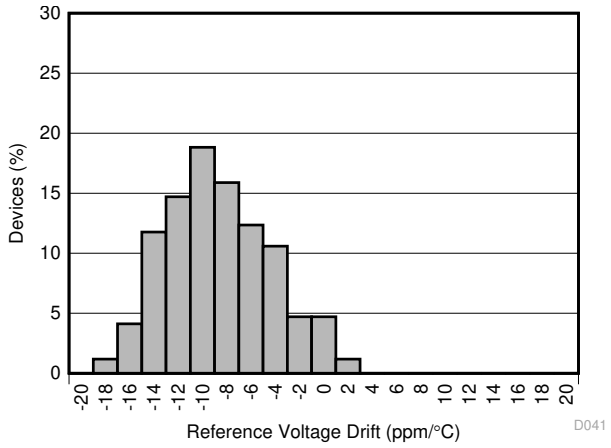



**Figure 5-53. Reference Voltage Drift Histogram**

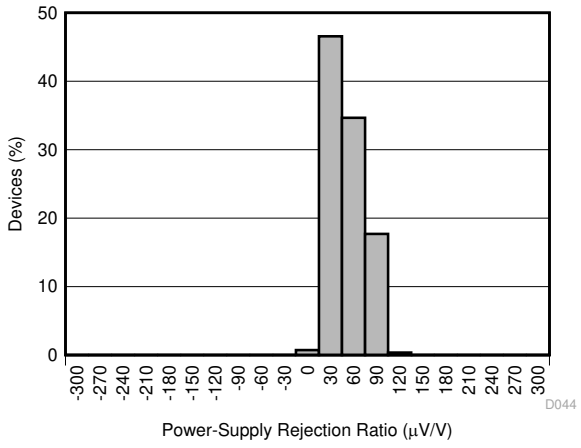



**Figure 5-54. Reference Voltage PSRR Histogram**


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## 5.6 Typical Characteristics (continued)

at VDD = 5V and $T_A$ = 25°C (unless otherwise noted)

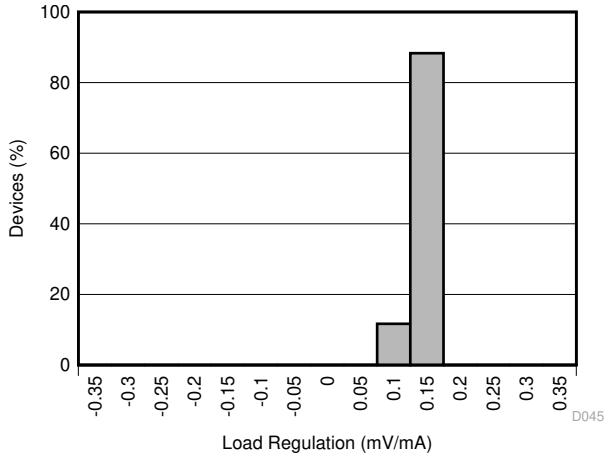



**Figure 5-55. Reference Voltage Load Regulation Histogram**

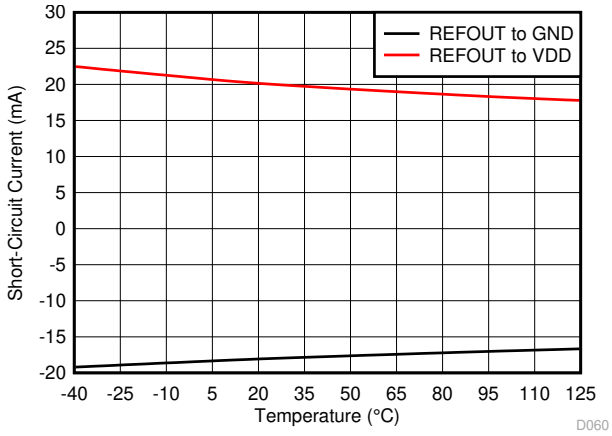



**Figure 5-56. Reference Short-Circuit Current vs Temperature**

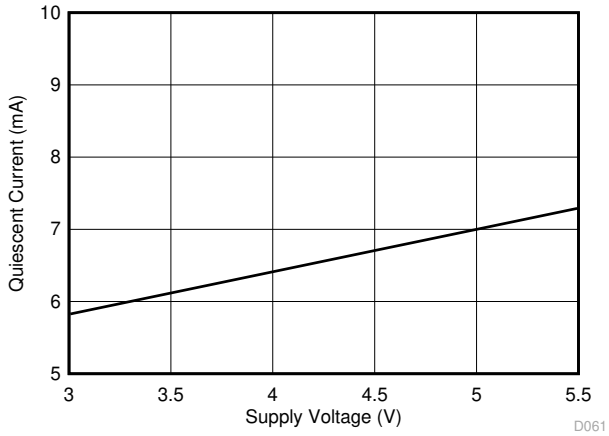



**Figure 5-57. Quiescent Current vs Supply Voltage**

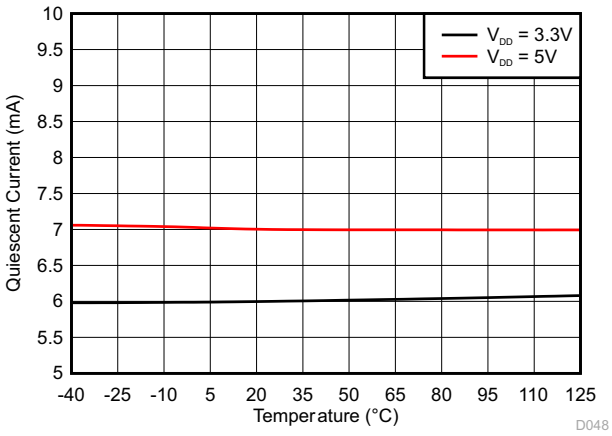



**Figure 5-58. Quiescent Current vs Temperature**

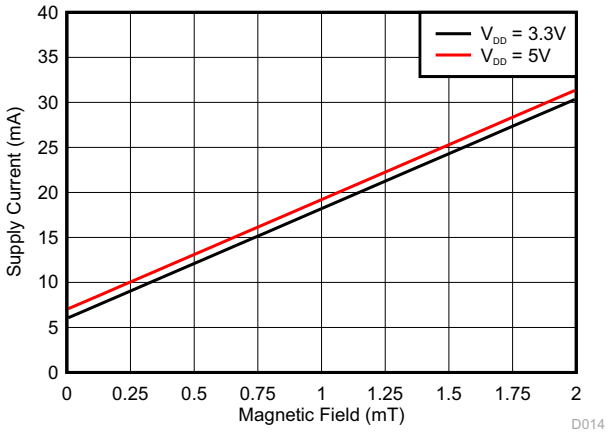



**Figure 5-59. Supply Current vs Magnetic Field**

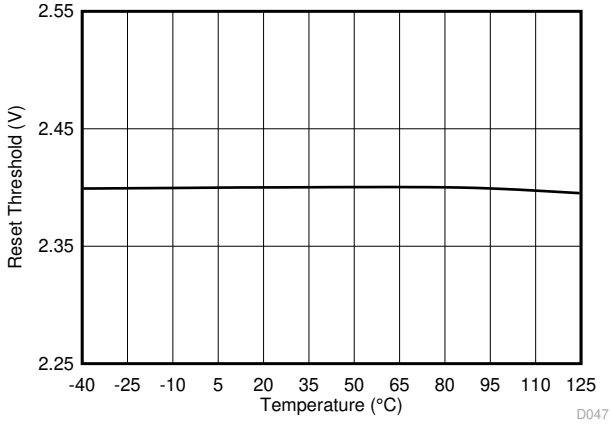



**Figure 5-60. Power-On Reset Threshold vs Temperature**

# 6 Detailed Description

## 6.1 Overview

Magnetic sensors are used in a broad range of applications, such as position, indirect ac and dc current, or torque measurement. Hall-effect sensors are most commonly used in magnetic field sensing, but offset, noise, gain variation, and nonlinearity limit the achievable resolution and accuracy of the system. Fluxgate sensors offer significantly higher sensitivity, lower drift, lower noise, high linearity, and enable up to 1000-times better measurement accuracy.

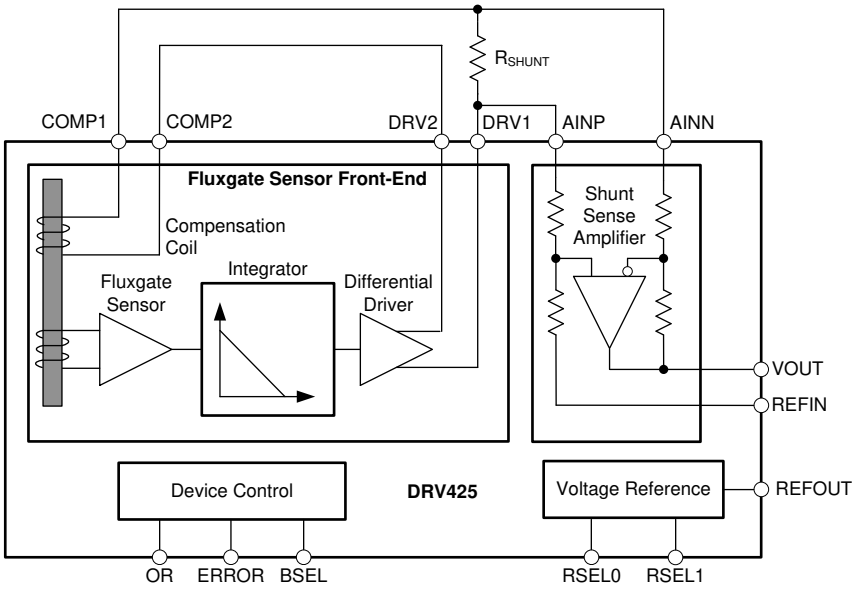
As shown in the *Functional Block Diagram* section, the DRV425 consists of a magnetic fluxgate sensor with the necessary sensor conditioning and compensation coil to internally close the control loop. The fluxgate sensor is repeatedly driven in and out of saturation, and supports hysteresis-free operation with excellent accuracy. The internal compensation coil provides stable gain and high linearity.

The magnetic field, B, is detected by the internal fluxgate sensor in the DRV425. The device integrates the sensor output to provide high-loop gain. The integrator output connects to the built-in differential driver that drives an opposing compensation current through the internal compensation coil. The compensation coil generates an opposite magnetic field that brings the original magnetic field at the sensor back to zero.

The compensation current is proportional to the external magnetic field, with a value of 12.2mA/mT. This compensation current generates a voltage drop across an external shunt resistor, $R_{SHUNT}$. An integrated difference amplifier with a fixed gain of 4V/V measures this voltage and generates an output voltage that is referenced to REFIN, and is proportional to the magnetic field. The value of the output voltage at the VOUT pin ($V_{VOUT}$) is calculated using Equation 1:

$$V_{VOUT}\,[V] = B \times G \times R_{SHUNT} \times G_{AMP} = B\,[mT] \times 12.2mA/mT \times R_{SHUNT}\,[\Omega] \times 4\,[V/V] \quad (1)$$

## 6.2 Functional Block Diagram





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## 6.3 Feature Description

### *6.3.1 Fluxgate Sensor Front-End*

The following sections describe the functional blocks and features of the integrated fluxgate sensor front-end.

#### 6.3.1.1 Fluxgate Sensor

The fluxgate sensor of the DRV425 is uniquely designed for high-performance magnetic-field sensors because of the high sensitivity, low noise, and low offset of the sensor. The fluxgate principle relies on repeatedly driving the sensor in and out of saturation; therefore, the sensor is free of any significant magnetic hysteresis. The feedback loop accurately drives a compensation current through the integrated compensation coil and drives the magnetic field at the sensor back to zero. This approach supports excellent gain stability and high linearity of the measurement.

The device package is free of any ferromagnetic materials to prevent magnetization by external fields and to obtain accurate and hysteresis-free operation. Select materials that cannot be magnetized for the printed circuit board (PCB) and passive components in the direct vicinity of the DRV425; see the *Layout Guidelines* section for more details.

The orientation and the sensitivity axis of the fluxgate sensor is indicated by a dashed line on the top of the package, as shown in Figure 6-1. The figure also shows the location of the sensor inside the package.

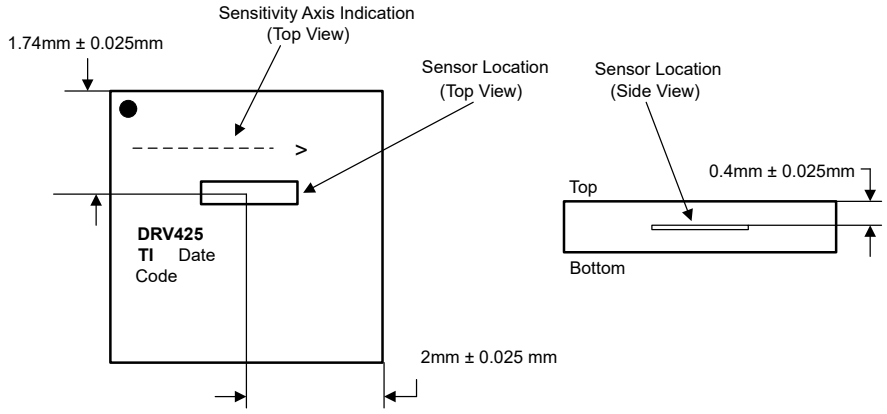



**Figure 6-1. Magnetic Sensitivity Direction of the Integrated Fluxgate Sensor**

The sensitivity of the fluxgate sensor is a vector function of the sensitivity axis and the magnitude of the magnetic field along that axis. Figure 6-2 shows the output of the DRV425 versus the angle of the device orientation relative to a constant magnetic field.

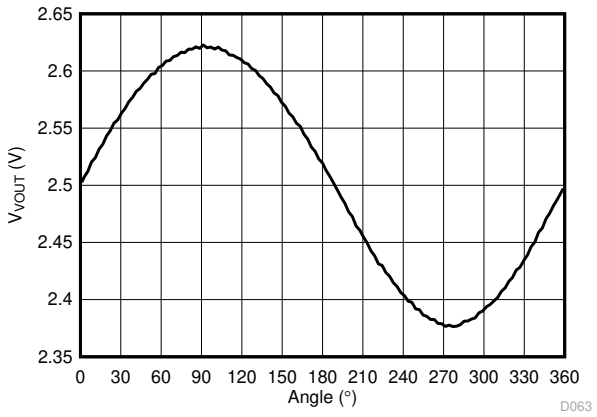



**Figure 6-2. Device Output vs. Magnetic Field Orientation**

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#### 6.3.1.2 Bandwidth

The small-signal bandwidth of the DRV425 is determined by the behavior of the compensation loop versus frequency. The implemented integrator limits the bandwidth of the loop to provide a stable response. Use digital input pin BSEL to select the bandwidth. With a shunt resistor of 22Ω and BSEL = 0, the bandwidth is 32kHz; for BSEL = 1, the bandwidth is 47kHz.

The shunt resistor and the compensation coil resistance form a voltage divider; therefore, to reduce the bandwidth, increase the value of the shunt resistor. To calculate the reduced bandwidth (BW), use Equation 2:

$$BW = \frac{R_{COIL} + 22\Omega}{R_{COIL} + R_{SHUNT}} \times BW_{22\Omega} = \frac{122\Omega}{100\Omega + R_{SHUNT}} \times BW_{22\Omega} \quad (2)$$

where

- $R_{COIL}$ = internal compensation coil resistance (100Ω).
- $R_{SHUNT}$ = external shunt resistance.
- $BW_{22\Omega}$ = sensor bandwidth with $R_{SHUNT}$ = 22Ω (depending on the BSEL setting).

The bandwidth for a given shunt resistor value can also be calculated using the *DRV425 System Parameter Calculator*. For large magnetic fields (B > 500µT), the effective bandwidth of the sensor is limited by fluxgate saturation effects. For a magnetic signal with a 2mT amplitude, the large-signal bandwidth is 10kHz with BSEL = 0, or 15kHz with BSEL = 1.

Although the analog output responds slowly to large fields, a magnetic field with a magnitude ≥ 1.6mT beyond the measurement range of the DRV425 triggers the $\overline{ERROR}$ pin within 4µs to 6µs. See the *Magnetic Field Range, Overrange Indicator, and Error Flag* section for more details.

#### 6.3.1.3 Differential Driver for the Internal Compensation Coil

The differential compensation coil driver provides the current for the internal compensation coil at the DRV1 and DRV2 pins. The driver is capable of sourcing up to ±250mA with a 5V supply, or up to ±150mA with a 3.3V supply. The current capability is not internally limited. The actual value of the compensation coil current depends on the magnetic field strength, and is limited by the sum of the resistance of the internal compensation coil and the external shunt resistor value. The internal compensation coil resistance depends on temperature (see Figure 5-17), and this dependency must be taken into account when designing the system. Select the value of the shunt resistor to avoid $\overline{OR}$ pin trip levels in normal operation.

The common-mode voltage of the compensation coil driver outputs is set by the RSEL pins; see the *Section 6.3.3* section. Thus, the common-mode voltage of the shunt-sense amplifier is matched if the internal reference is used.

Consider the polarity of the compensation coil connection to the output of the compensation coil driver. If the polarity is incorrect, then the driver output drives to the power-supply rails, even at low primary-current levels. In this case, interchange the connection of the DRV1 and DRV2 pins to the compensation coil.


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#### 6.3.1.4 Magnetic Field Range, Overrange Indicator, and Error Flag

The measurement range of the DRV425 is determined by the amount of current driven into the compensation coil and the output voltage range of the shunt-sense amplifier. The maximum compensation current is limited by the supply voltage and the series resistance of the compensation coil and the shunt.

The magnetic field range is adjusted with the external shunt resistor. The *DRV425 System Parameter Calculator* provides the maximum shunt resistor values depending on the supply voltage (VDD) and the selected reference voltage ($V_{REFIN}$) for various magnetic field ranges.

For proper operation at a maximum field ($B_{MAX}$), choose a shunt resistor ($R_{SHUNT}$) using Equation 3:

$$R_{SHUNT} \leq \frac{\min((VDD - V_{REFIN}),\ V_{REFIN}) - 0.085V}{B_{MAX} \times 12.2A/T \times 4V/V} \quad (3)$$

where

- VDD = minimum supply voltage of the DRV425 (V).
- $V_{REFIN}$ = common-mode voltage of the shunt-sense amplifier (V).
- $B_{MAX}$ = desired magnetic field range (T).

Alternatively, to adjust the output voltage of the DRV425 for a desired maximum voltage ($V_{VOUTMAX}$), use Equation 4:

$$R_{SHUNT} \leq \frac{V_{VOUTMAX} - V_{REFIN}}{B_{MAX} \times 12.2A/T \times 4V/V} \quad (4)$$

where

- $V_{VOUTMAX}$ = desired maximum output voltage at VOUT pin (V).
- $B_{MAX}$ = desired magnetic field range (T).

To avoid railing of the compensation coil driver, make sure that Equation 5 is fulfilled:

$$\frac{B_{MAX} \times (R_{COIL} + R_{SHUNT}) \times 12.2A/T}{2} + 0.1V \leq \min((VDD - V_{REFIN}),\ V_{REFIN}) \quad (5)$$

where

- $B_{MAX}$ = desired magnetic field range (T).
- $R_{COIL}$ = compensation coil resistance (Ω).
- VDD = minimum supply voltage of the DRV425 (V).
- $V_{REFIN}$ = selected internal reference voltage value (V).

The *DRV425 System Parameter Calculator* is designed to assist with selecting the system parameters.

The DRV425 offers two diagnostic output pins to detect large fields that exceed the measurement range of the sensor: the overrange indicator (OR) and the ERROR flag.

In normal operation, the DRV425 sensor feedback loop compensates the magnetic field inside the fluxgate to zero. Therefore, a large field inside the fluxgate indicates that the feedback loop is not properly working, and the sensor output is invalid. To detect this condition, the $\overline{\text{ERROR}}$ pin is pulled low if the internal field exceeds 1.6mT. The $\overline{\text{ERROR}}$ output is suppressed for 4µs to 6µs to prevent an undesired reaction to transients or noise. For static and slowly varying ambient fields, the $\overline{\text{ERROR}}$ pin triggers when the ambient field exceeds the sensor measurement range by more than 1.6mT. For dynamic magnetic fields that exceed the sensor bandwidth as specified in the *Section 5* section, the feedback loop response is too slow to accurately compensate the internal field to zero. Therefore, high-frequency fields can trigger the $\overline{\text{ERROR}}$ pin, even if the ambient field does not exceed the measurement range by 1.6mT.

In addition, the active-low overrange pin ( $\overline{\text{OR}}$) indicates railing of the output of the shunt-sense amplifier. The $\overline{\text{OR}}$ output is suppressed for 2.5µs to 3.5µs to prevent an undesired reaction to transients or noise. The $\overline{\text{OR}}$ pin

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trip level refers to the output voltage value of the shunt-sense amplifier, as specified in the *Section 5* section. Use Equation 3 and Equation 4 to adjust the $\overline{OR}$ pin behavior to the specific system-level requirements.

Both the $\overline{ERROR}$ and $\overline{OR}$ pins are open-drain outputs that require an external pullup resistor. If desired, connect both pins together with a single pullup resistor to provide a single diagnostic flag.

Based on the *DRV425 System Parameter Calculator*, for a design for a ±2mT magnetic field input range with a supply of 5V (±5%), a shunt resistor value of 22Ω is selected. Figure 6-3 shows the status of the diagnostic flags in the resulting three operation ranges.

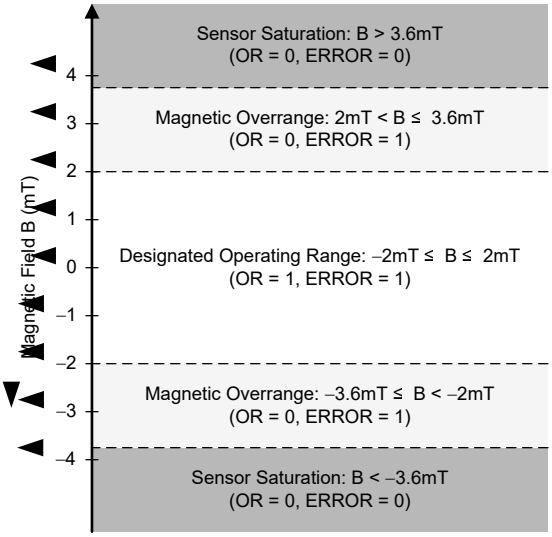



**Figure 6-3. Magnetic Field Range of the DRV425 (VDD = 5V and $R_{SHUNT}$ = 22Ω)**

With the proper $R_{SHUNT}$ value, the differential amplifier output rails and activates the overrange flag (OR = 0) when the magnetic field exceeds the designated operating range. For fields that exceed the measurement range of the DRV425 by ≥ 1.6mT, the fluxgate is saturated and the $\overline{ERROR}$ pin is pulled low. In this condition, the fluxgate sensor does not provide a valid output value; therefore, the output VOUT of the DRV425 must be ignored. In applications where the $\overline{ERROR}$ pin cannot be separately monitored, combine the VOUT and ERROR outputs as shown in Figure 6-4. This method indicates that a magnetic field is outside of the sensor range by pulling the device output to ground.

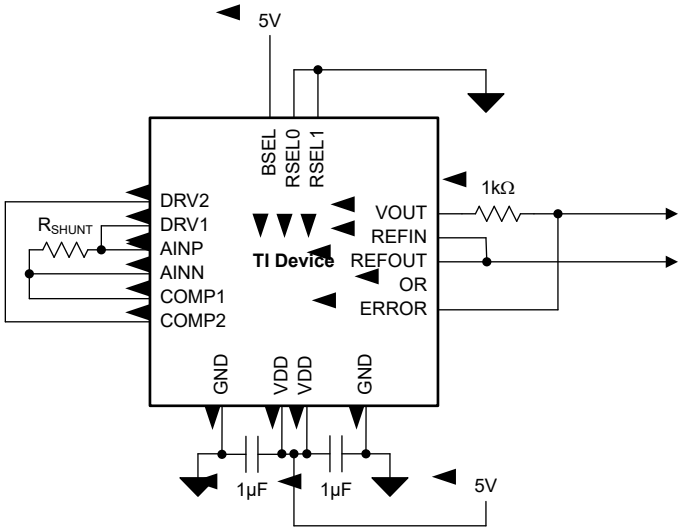



**Figure 6-4. Field Overrange Detection Using a Combined VOUT and $\overline{ERROR}$ Pin**


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### 6.3.2 Shunt-Sense Amplifier

The compensation coil current creates a voltage drop across the external shunt resistor, $R_{SHUNT}$. The internal differential amplifier senses this voltage drop. This differential amplifier offers wide bandwidth and a high slew rate. Excellent dc stability and accuracy result from a chopping technique. The voltage gain is 4V/V, set by precisely matched and thermally stable internal resistors.

Both the AINN and AINP differential amplifier inputs are connected to the external shunt resistor. This shunt resistor, in series with the internal 10kΩ input resistors of the shunt-sense amplifier, causes an additional gain error. Therefore, for best common-mode rejection performance, place a dummy shunt resistor ($R_5$) with a value higher than the shunt resistor in series with the REFIN pin to restore the matching of both resistor dividers, as shown in Figure 6-5.

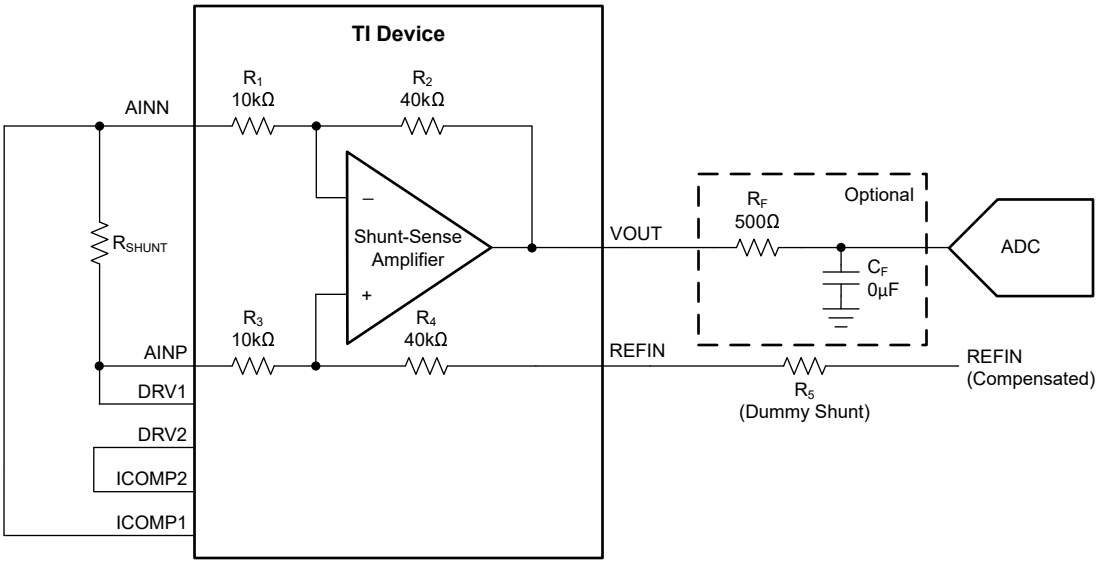



**Figure 6-5. Internal Difference Amplifier With an Example of a Decoupling Filter**

For an overall gain of 4V/V, calculate the value of $R_5$ using Equation 6:

$$4 = \frac{R_2}{R_1} = \frac{R_4 + R_5}{R_{SHUNT} + R_3} \quad (6)$$

where:

- $R_2 / R_1 = R_4 / R_3 = 4$.
- $R_5 = R_{SHUNT} \times 4$.

If the input signal is large, the amplifier output drives close to the supply rails. The amplifier output is able to drive the input of a successive approximation register (SAR) analog-to-digital converter (ADC). For best performance, add an RC low-pass filter stage between the shunt-sense amplifier output and the ADC input. This filter limits the noise bandwidth, and decouples the high-frequency sampling noise of the ADC input from the amplifier output. For filter resistor $R_F$ and filter capacitor $C_F$ values, see the specific converter recommendations in the respective product data sheet.

The shunt-sense amplifier output drives 100pF directly, and shows a 50% overshoot with a 1nF capacitance. Filter resistor $R_F$ extends the capacitive load range. With an $R_F$ of only 20Ω, the load capacitor must be either less than 1nF or more than 33nF to avoid overshoot; with an $R_F$ of 50Ω, this transient area is avoided.

Reference input REFIN is the common-mode voltage node for the output signal VOUT. To use the internal voltage reference of the DRV425, connect the REFIN pin to the reference output REFOUT. To avoid mismatch errors, use the same reference voltage for REFIN and the ADC. Alternatively, use an ADC with a pseudo-

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differential input, with the positive input of the ADC connected to VOUT, and the negative input connected to REFIN of the device.

### 6.3.3 Voltage Reference

The internal precision voltage reference circuit offers low-drift performance at the REFOUT output pin, and is used for internal biasing. The reference output is intended to be the common-mode voltage of the output (the VOUT pin) to provide a bipolar signal swing. This low-impedance output tolerates sink and source currents of ±5mA. However, fast load transients can generate ringing on this line. A small series resistor of a few ohms improves the response, particularly for capacitive loads equal to or greater than 1µF.

To adjust the value of the voltage reference output to the power supply of the DRV425, use mode selection pins RSEL0 and RSEL1, as shown in Table 6-1.

**Table 6-1. Reference Output Voltage Selection**

| MODE | RSEL1 | RSEL0 | DESCRIPTION |
|---|---|---|---|
| $V_{REFOUT}$ = 2.5V | 0 | 0 | Use with a sensor module supply of 5V |
| $V_{REFOUT}$ = 1.65V | 0 | 1 | Use with a sensor module supply of 3.3V |
| Ratiometric output | 1 | x | Provides an output centered on VDD / 2 |

In ratiometric output mode, an internal resistor divider divides the power-supply voltage by a factor of two.

### 6.3.4 Low-Power Operation

In applications with low-bandwidth or low sample-rate requirements, significantly reduce the average power dissipation of the DRV425 by powering down the device between measurements. The DRV425 requires 300µs to fully settle the analog output VOUT, as shown in Figure 6-6. To minimize power dissipation, power down the device immediately after the ADC acquires the sample.

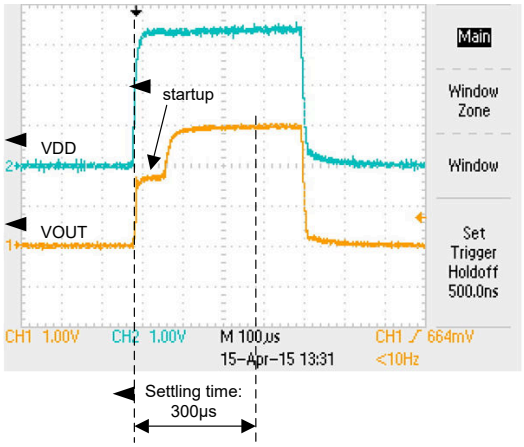



**Figure 6-6. Settling Time of the DRV425 VOUT Output**

## 6.4 Device Functional Modes

The DRV425 is operational when the power supply VDD is applied, as specified in the *Specifications* section. The DRV425 has no additional functional modes.


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# 7 Application and Implementation

**Note**

Information in the following applications sections is not part of the TI component specification, and TI does not warrant its accuracy or completeness. TI's customers are responsible for determining suitability of components for their purposes, as well as validating and testing their design implementation to confirm system functionality.

## 7.1 Application Information

The DRV425 is a high-sensitivity and high-performance magnetic-field sensor. The analog output of the DRV425 can be processed by a 12-bit to 16-bit analog to digital converter (ADC). The following sections show application design examples.

## 7.2 Typical Applications

### *7.2.1 Linear Position Sensing*

The high sensitivity of the fluxgate sensor, combined with the high linearity of the compensation loop and low noise of the DRV425, make the device an excellent choice for high-performance linear-position sense applications. A typical schematic of such a 5V application using an internal 2.5V reference is shown in Figure 7-1.

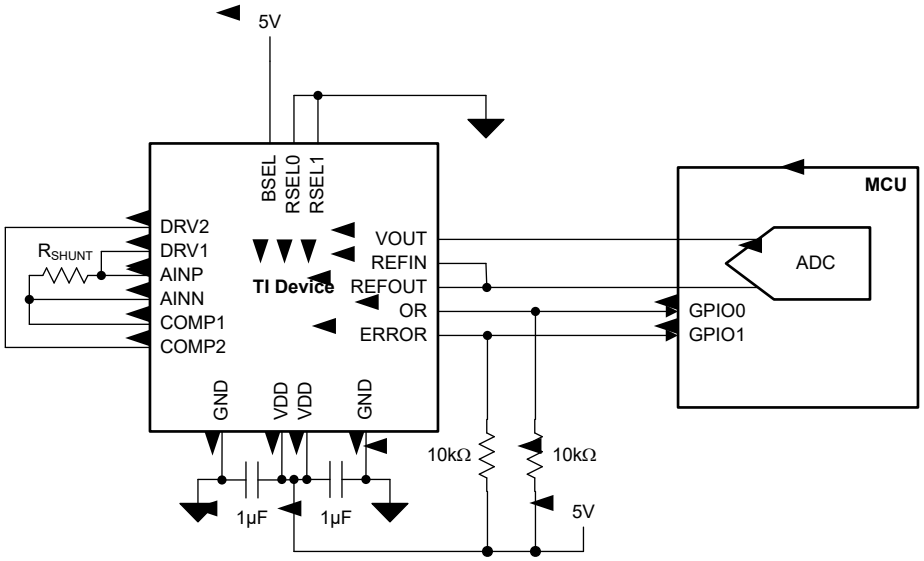



**Figure 7-1. Linear-Position Sensing**

#### 7.2.1.1 Design Requirements

For the example shown in Figure 7-1, use the parameters listed in Table 7-1 as a starting point of the design.

**Table 7-1. Design Parameters**

| DESIGN PARAMETER | EXAMPLE VALUE |
|---|---|
| Magnetic field range | VDD = 5V: ±2mT (max)<br>VDD = 3.3V: ±1.3mT (max) |
| Supply voltage, VDD | 3.0V to 5.5V |
| Reference voltage, $V_{REFIN}$ | Range: GND to VDD<br>If an internal reference is used: 2.5V, 1.65V, or VDD / 2 |
| Shunt resistor, $R_{SHUNT}$ | Depends on the desired magnetic field range, reference, and supply voltage; see the *DRV425 System Parameter Calculator* for details. |

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#### 7.2.1.2 Detailed Design Procedure

Use the following procedure to create a design for a linear-position sensor based on the DRV425:

1. Select the proper supply voltage, VDD, to support the desired magnetic field range (see Table 7-1 for reference).
2. Select the proper reference voltage, $V_{REFIN}$, to support the desired magnetic field range and to match the input voltage specifications of the desired ADC.
3. Use the *RangeCalculator* tab in the *DRV425 System Parameter Calculator* to select the proper shunt resistor value of $R_{SHUNT}$.
4. The sensitivity drift performance of a DRV425- based linear position sensor is dominated by the temperature coefficient of the external shunt resistor. Select a low-drift shunt resistor for best sensor performance.
5. Use the *Problems Detected Table in DRV425 System Parameters* tab in the *DRV425 System Parameter Calculator* to verify the system response.

The amplitude of the magnetic field is a function of distance to, and the shape of, the magnet, as shown in Figure 7-3. If the magnetic field to be measured exceeds 3.6mT, see the magnet data sheet to calculate the appropriate minimum distance to the DRV425 to avoid saturating the fluxgate sensor.

The high sensitivity of the DRV425 can require shielding of the sensing area to avoid influence of undesired magnetic field sources (such as the earth magnetic field). Alternatively, an additional DRV425 can be used to perform difference measurement to cancel the influence of a static magnetic field source, as shown in Figure 7-2. Figure 7-4 shows the differential voltage generated by two DRV425 devices in such a circuit.

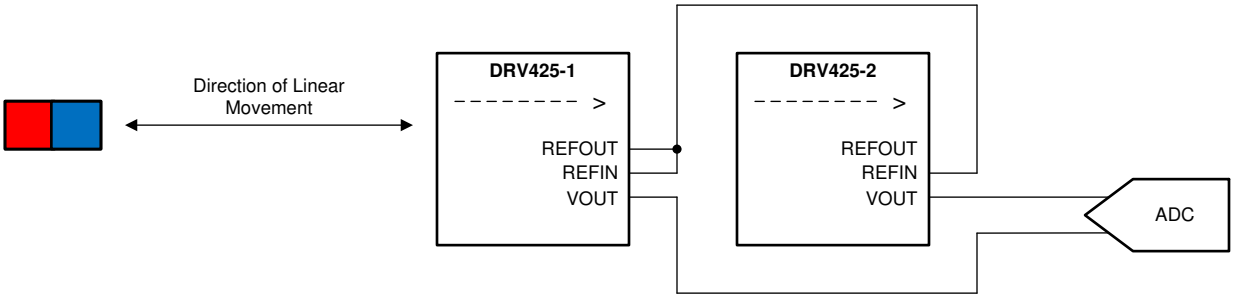



**Figure 7-2. Differential Linear-Position Sensing Using Two DRV425 Devices**

#### 7.2.1.3 Application Curves

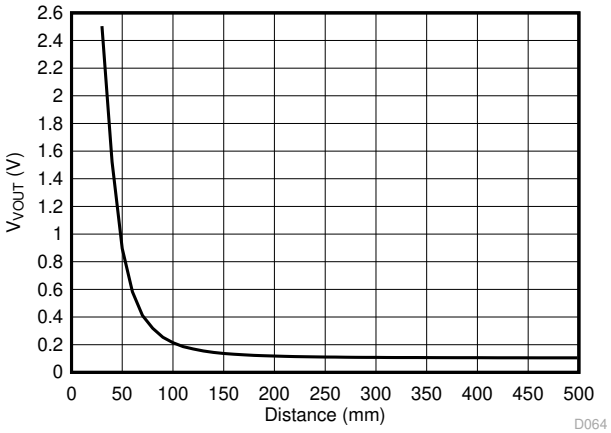



**Figure 7-3. Analog Output Voltage of the DRV425 vs Distance to the Magnet**

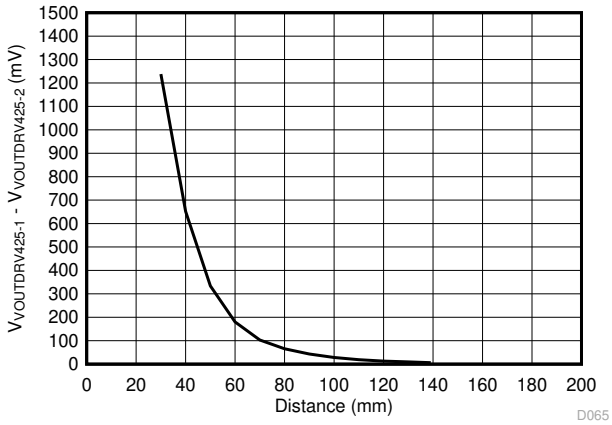



**Figure 7-4. Difference Between Two DRV425 Outputs vs Distance to the Magnet**


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### 7.2.2 Current Sensing in Busbars

In existing applications that use busbars for power distribution, closed-loop current modules are typically used to accurately measure and control the current. These modules are typically bulky because of the required large magnetic core. Additionally, because the compensation current generated inside the module is proportional to the typically high busbar current, the power dissipation of this design is typically as high as several watts.

Figure 7-5 shows an alternative approach with two DRV425 devices. If a hole is drilled in the middle of the busbar, the current is split in two equal parts that generate magnetic field gradients with opposite directions inside the hole. These magnetic fields are termed $B_R$ and $B_L$ in Figure 7-6. The opposite fields cancel each other out in the middle of the hole. The high sensitivity and linearity of two DRV425 devices positioned at the same distance from the middle of the hole allow the small opposite fields to be sensed and the current measured with high-accuracy levels. The differential measurement rejects outside fields that generate a common-mode error that is subtracted at the output.

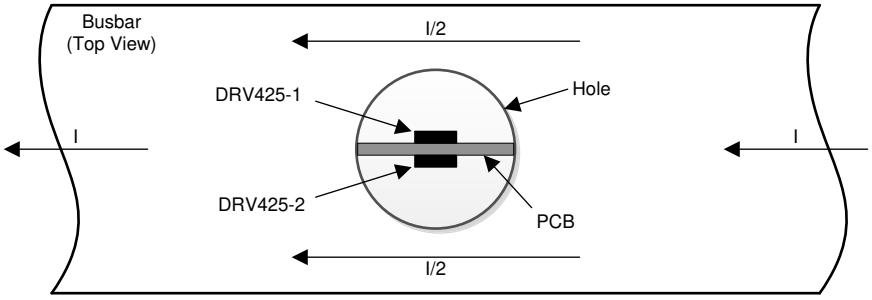



**Figure 7-5. Current Sensing in Busbars**

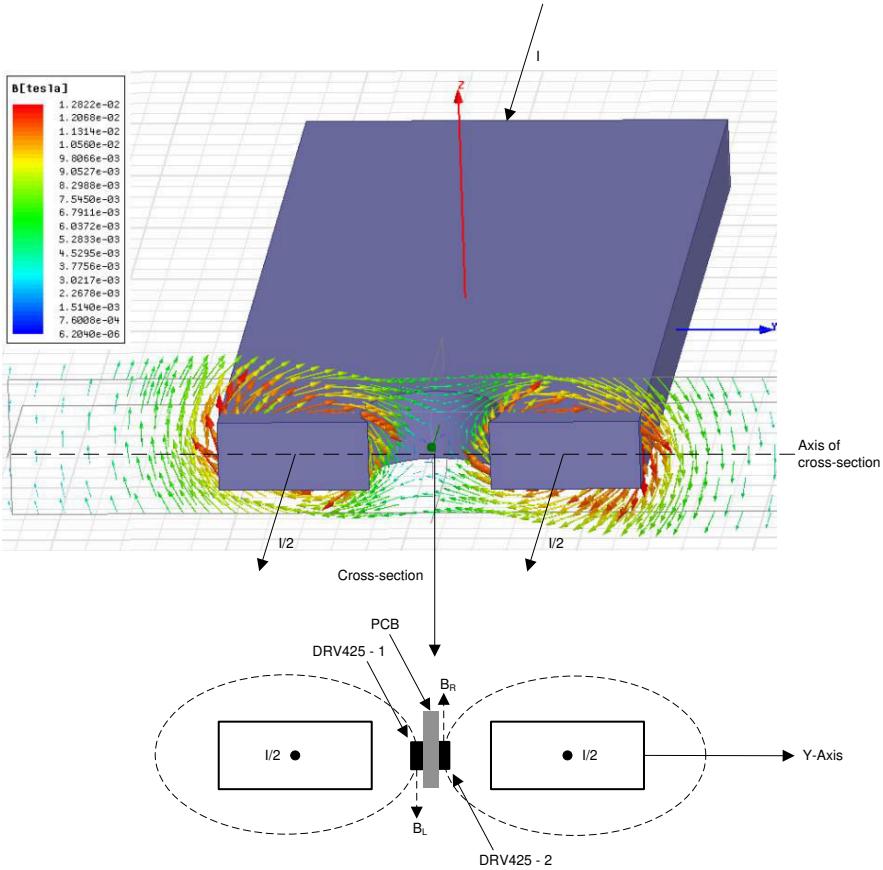



**Figure 7-6. Magnetic Field Distribution Inside a Busbar Hole**

#### 7.2.2.1 Design Requirements

To measure the field gradient in the busbar, two DRV425 sensors are placed inside the hole at a well-defined distance by mounting them on opposite sides of a PCB that is inserted in the hole. The measurement range and resolution of this design depends on the following factors:

- Busbar geometry: a wider busbar means a larger measurement range and lower resolution.
- Size of the hole: a larger diameter means a larger measurement range and lower resolution.
- Distance between the two DRV425 sensors: a smaller distance increases the measurement range and resolution.

Each of these factors can be optimized to create the desired measurement range for a particular application. Measurement ranges of ±250A to ±1500A are achievable with this approach. Larger currents are supported with large busbar structures and minimized distance between the two DRV425 sensors. Use the parameters listed in Table 7-2 as a starting point of the design.

**Table 7-2. Design Parameters**

| DESIGN PARAMETER | EXAMPLE VALUE |
|---|---|
| Current range | Up to ±1500A |
| Supply voltage, VDD | 3.0V to 5.5V |
| Reference voltage, $V_{REFIN}$ | VDD / 2 |

#### 7.2.2.2 Detailed Design Procedure

Figure 7-7 shows the schematic diagram of a differential gradient field measurement circuit.

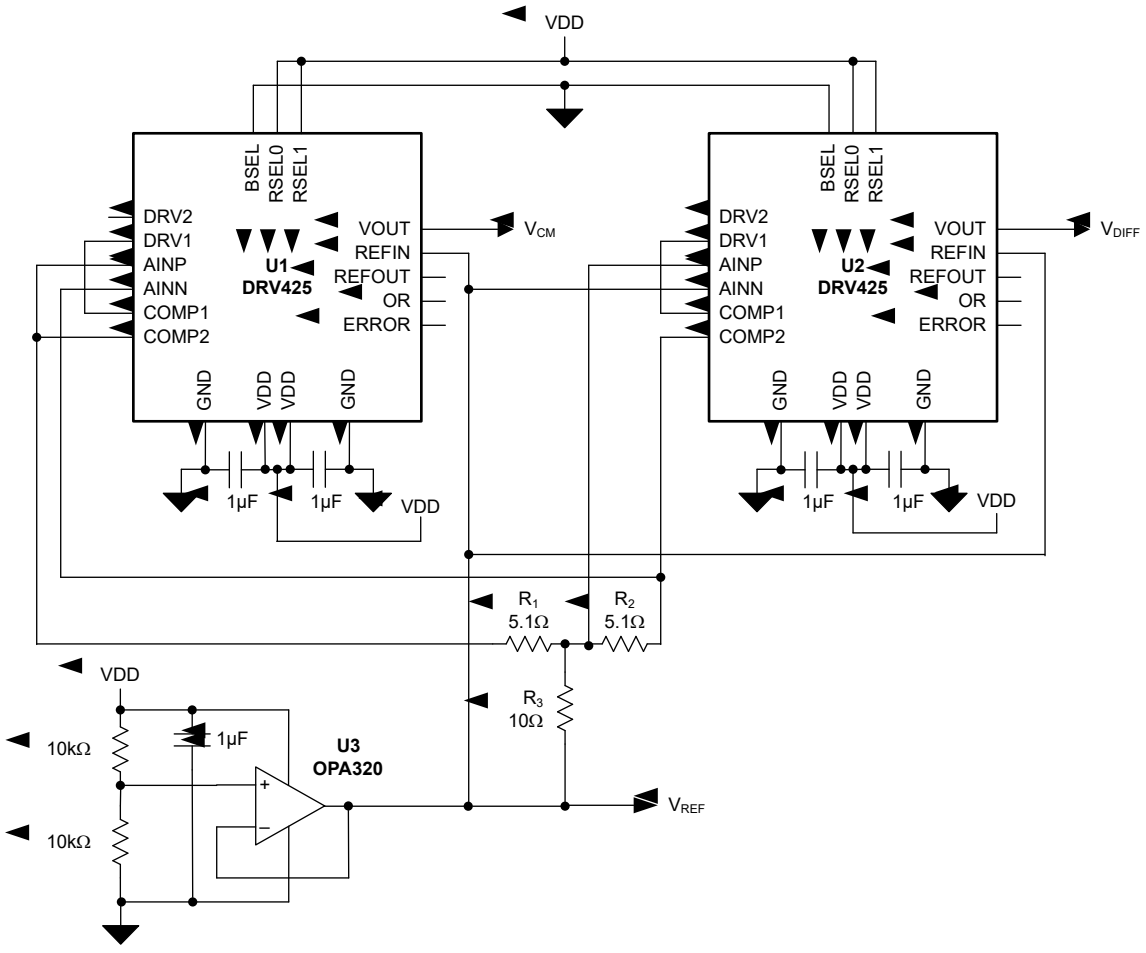



**Figure 7-7. Busbar Current-Sensing Circuit**


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In Figure 7-7, the feedback loops of both DRV425 sensors are combined to directly produce differential output $V_{DIFF}$ that is proportional to the sensed magnetic field difference inside the busbar hole. Both compensation coils are connected in series and are driven from a single side of the compensation coil driver (the DRV1 pins of each DRV425). Therefore, both driver stages make sure that a current proportional to the magnetic fields $B_R$ and $B_L$ is driven through the respective compensation coil. The difference in current through both compensation coils, and thus the difference field between the sensors, flows through resistor $R_3$, and is sensed by the shunt-sense amplifier of U2. The current proportional to the common-mode field inside the busbar hole flows through $R_1$ and $R_2$, and is sensed by the shunt-sense amplifier of U1.

Use the output $V_{CM}$ to verify that the sensors are correctly positioned in the busbar hole with the following steps:

1. Measure $V_{CM}$ with no current flow through the busbar and the PCB in the middle of the busbar hole. This value is the offset voltage $V_{OFFSET}$. The value of $V_{OFFSET}$ only depends on stray fields and varies little with the absolute position of the sensors.
2. Apply current through the busbar and move the PCB along the y-axis in the busbar hole, as shown in Figure 7-6. The PCB is in the center of the hole if $V_{CM} = V_{OFFSET}$.

The sensitivity drift performance of the circuit shown in Figure 7-7 is dominated by the temperature coefficient of the external resistors $R_1$, $R_2$, and $R_3$. Select low-drift resistors for best sensor performance. For overall system error calculation, also consider the affect of thermal expansion on the PCB and busbar.

The internal voltage reference of the DRV425 cannot be used in this application because of the limited driver capability. The OPA320 (U3) is a low-noise operational amplifier with a short-circuit current capability of ±65mA, and is used to support the required compensation current.

The advantage of this design is the simplicity: the currents are subtracted by the two DRV425 devices without additional components. The series connection of the compensation coils halves the voltage swing, and reduces the measurement range of the sensors also by 50%. If a larger sensing range is required, operate the two sensors independently, and use a differential amplifier or ADC to subtract both voltage outputs (VOUT).

Use the $\overline{\text{ERROR}}$ outputs for fast overcurrent detection on the system level.

#### 7.2.2.3 Application Curves

Figure 7-8 and Figure 7-9 show the measurement results on a 16mm wide and 6mm thick copper busbar with a 12mm hole diameter using the circuit shown in Figure 7-7. The two DRV425 devices are placed at a distance of 1mm from each other on opposite sides of the PCB. The measurement range is ±500A; measurement results are limited by test setup. Independent operation of the two DRV425 sensors increases the measurement range to ±1000A with the same busbar geometry.

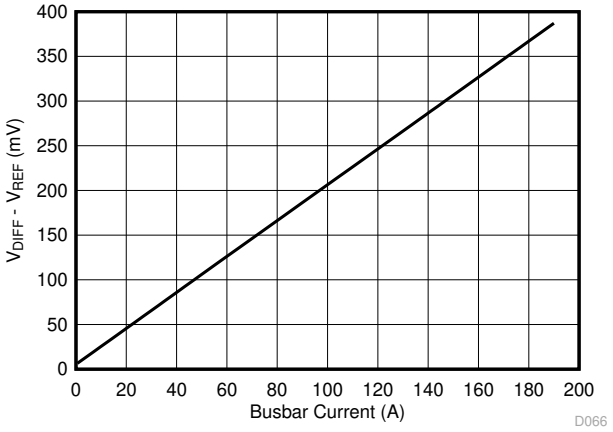



**Figure 7-8. Analog Output Voltage vs Busbar Current**

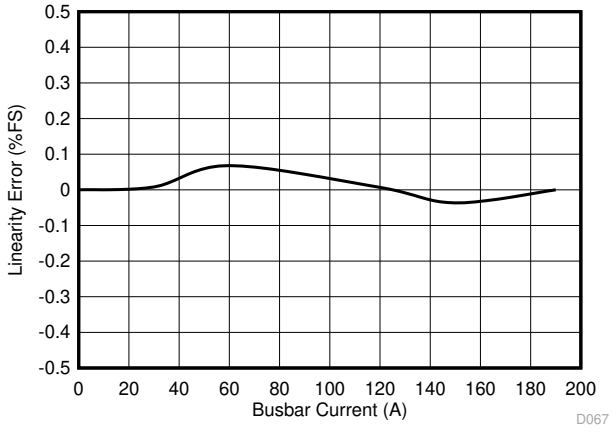



**Figure 7-9. Linearity Error vs Busbar Current**

## 7.3 Power Supply Recommendations

### 7.3.1 Power Supply Decoupling

Decouple both VDD pins of the DRV425 with 1µF, X7R-type ceramic capacitors to the adjacent GND pin, as illustrated in Figure 7-10. For best performance, place both decoupling capacitors as possible close to the related power-supply pins. Connect these capacitors to the power-supply source in a way that allows the current to flow through the pads of the decoupling capacitors.

### 7.3.2 Power-On Start-Up and Brownout

Power-on is detected when the supply voltage exceeds 2.4V at the VDD pin. At this point, the DRV425 initiates the following start-up sequence:

1. Digital logic starts up and waits for 26µs for the supply to settle.
2. The fluxgate sensor powers up.
3. The compensation loop is active 70µs after the supply voltage exceeds 2.4V.

During this startup sequence, the DRV1 and DRV2 outputs are pulled low to prevent undesired signals on the compensation coil and the $\overline{\text{ERROR}}$ pin is asserted low.

The DRV425 tests for low supply voltages with a brownout-voltage level of 2.4V. Use a power-supply source capable of supporting large current pulses driven by the DRV425, and low-ESR bypass capacitors for a stable supply voltage in the system. A supply drop to less than 2.4V that lasts longer than 20µs generates a power-on reset; the device ignores shorter voltage drops. A voltage drop on the VDD pin to below 1.8V immediately initiates a power-on reset. After the power supply returns to 2.4V, the device initiates a start-up cycle.

### 7.3.3 Power Dissipation

The thermally-enhanced, WQFN package with thermal pad reduces the thermal impedance from junction to case. This package has a downset leadframe to which the die is mounted. The leadframe has an exposed thermal pad on the underside of the package, and provides a good thermal path for heat dissipation.

The power dissipation on both linear outputs DRV1 and DRV2 is calculated with Equation 7:

$$P_{D(DRV)} = I_{DRV} \times (V_{DRV} - V_{SUPPLY}) \quad (7)$$

where

- $I_{DRV}$ = supply current as shown in Figure 5-59.
- $V_{DRV}$ = voltage potential on the DRV1 or DRV2 output pin.
- $V_{SUPPLY}$ = voltage potential closer to $V_{DRV}$: VDD or GND.

#### 7.3.3.1 Thermal Pad

Packages with an exposed thermal pad are specifically designed to provide excellent power dissipation, but board layout greatly influences the overall heat dissipation. Technical details are described in the *PowerPad Thermally Enhanced Package,* application report, available for download at www.ti.com.

## 7.4 Layout

### 7.4.1 Layout Guidelines

The unique, integrated fluxgate of the DRV425 has a very high sensitivity that enables designing a closed-loop magnetic-field sensor with best-in-class precision and linearity. Observe proper PCB layout techniques because any current-conducting wire in the direct vicinity of the DRV425 generates a magnetic field that can distort measurements. Common passive components and some PCB plating materials contain ferromagnetic materials that are magnetizable. For best performance, use the following layout guidelines:

- Route current-conducting wires in pairs: route a wire with an incoming supply current next to, or on top of, the return current path. The opposite magnetic field polarity of these connections cancel each other. To facilitate this layout approach, the DRV425 positive and negative supply pins are located adjacently.


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- Route the compensation coil connections close to each other as a pair to reduce coupling effects.
- Minimize the length of the compensation coil connections between the DRV1/2 and COMP1/2 pins.
- Route currents parallel to the fluxgate sensor sensitivity axis as illustrated in Figure 7-10. As a result, magnetic fields are perpendicular to the fluxgate sensitivity and have limited affect.
- Vertical current flow (for example, through vias) generates a field in the fluxgate-sensitive direction. Minimize the number of vias in the vicinity of the DRV425.
- Use passive components (for example, decoupling capacitors and the shunt resistor) that cannot be magnetized to prevent magnetic effects near the DRV425.
- Do not use PCB trace finishes with nickel-gold plating because of the potential for magnetization.
- Connect all GND pins to a local ground plane.

Ferrite beads in series with the power-supply connection reduce interaction with other circuits powered from the same supply voltage source. However, to prevent influence of the magnetic fields if ferrite beads are used, do not place them next to the DRV425.

The reference output (the REFOUT pin) refers to GND. Use a low-impedance and star-type connection to reduce the driver current and the fluxgate sensor current modulating the voltage drop on the ground track. The REFOUT and VOUT outputs are able to drive some capacitive load, but avoid large direct capacitive loading because of increased internal pulse currents. Given the wide bandwidth of the shunt-sense amplifier, isolate large capacitive loads with a small series resistor.

Solder the exposed thermal pad on the bottom of the package to the ground layer because the thermal pad is internally connected to the substrate that must be connected to the most-negative potential.

Figure 7-10 illustrates a generic layout example that highlights the placement of components that are critical to the DRV425 performance. For specific layout examples, see the *DRV425EVM* users guide.

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### 7.4.2 Layout Example

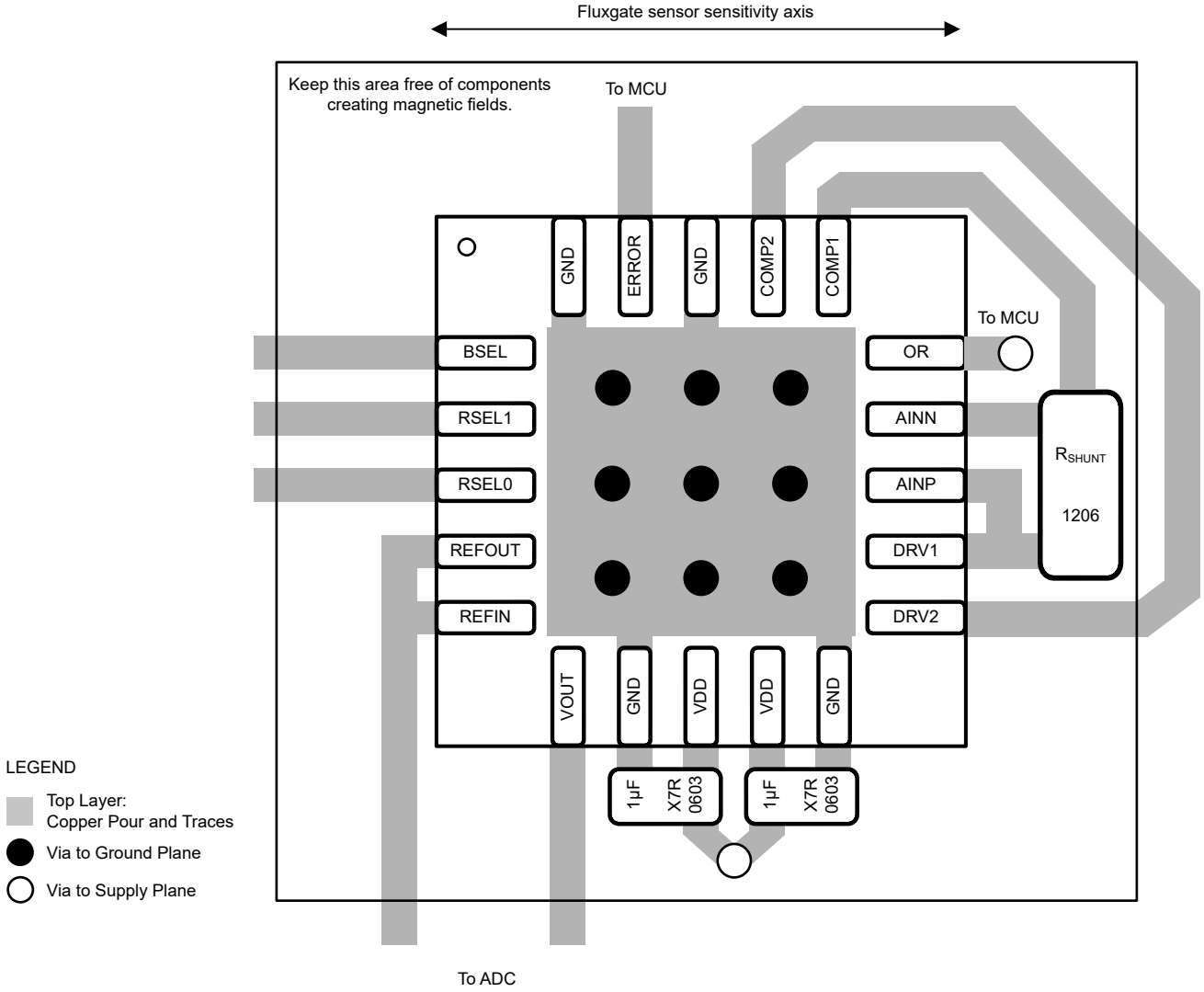



**Figure 7-10. Generic Layout Example (Top View)**


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# 8 Device and Documentation Support

## 8.1 Documentation Support

### *8.1.1 Related Documentation*

For related documentation see the following:

- Texas Instruments, *OPAx320 Precision, 20MHz, Low-Noise, Low-Power, RRIO, CMOS Op Amp With Shutdown data sheet*
- Texas Instruments, *DRV425EVM* user's guide
- Texas Instruments, *DRV425 System Parameter Calculator*
- Texas Instruments, *PowerPad Thermally Enhanced Package application note*
- Texas Instruments, *±100A Busbar Current Sensor Reference Design Using Open-Loop Fluxgate Sensors reference design TIPD205*
- Texas Instruments, *Bus Bar Theory of Operation application note*

## 8.2 Receiving Notification of Documentation Updates

To receive notification of documentation updates, navigate to the device product folder on ti.com. Click on *Notifications* to register and receive a weekly digest of any product information that has changed. For change details, review the revision history included in any revised document.

## 8.3 Support Resources

TI E2E™ support forums are an engineer's go-to source for fast, verified answers and design help — straight from the experts. Search existing answers or ask your own question to get the quick design help you need.

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## 8.5 Support Resources

## 8.6 Trademarks

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## 8.7 Electrostatic Discharge Caution



This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

## 8.8 Glossary

| | |
|---|---|
| TI Glossary | This glossary lists and explains terms, acronyms, and definitions. |

# 9 Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

| Changes from Revision A (March 2016) to Revision B (October 2025) | Page |
|---|---|
| • Changed instances of PowerPAD to thermal pad | 1 |
| • Changed PSSRAMP parameter min and max values from –50µV/V and 50µV/V to –86µV/V and 86µV/V, respectively | 5 |

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- Changed location of sensor details for clarity.....19
- Deleted "Permanently" from 2nd paragraph to clarify that the fluxgate is not permanently saturated..... 21
- Changed Figure 6-5 to show the corrected pin names.....23

**Changes from Revision * (October 2015) to Revision A (March 2016)** **Page**

- Changed broken links to working links..... 1
- Added last four Applications bullets .....1
- Changed device name in Figure 6-3 ..... 21

## 10 Mechanical, Packaging, and Orderable Information

The following pages include mechanical, packaging, and orderable information. This information is the most current data available for the designated devices. This data is subject to change without notice and revision of this document. For browser-based versions of this data sheet, refer to the left-hand navigation.


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## PACKAGING INFORMATION

| Orderable part number | Status (1) | Material type (2) | Package \| Pins | Package qty \| Carrier | RoHS (3) | Lead finish/ Ball material (4) | MSL rating/ Peak reflow (5) | Op temp (°C) | Part marking (6) |
|---|---|---|---|---|---|---|---|---|---|
| DRV425RTJR | Active | Production | QFN (RTJ) \| 20 | 3000 \| LARGE T&R | Yes | Call TI | Level-3-260C-168 HR | -40 to 125 | -----> DRV425 |
| DRV425RTJR.A | Active | Production | QFN (RTJ) \| 20 | 3000 \| LARGE T&R | Yes | Call TI | Level-3-260C-168 HR | -40 to 125 | -----> DRV425 |
| DRV425RTJT | Obsolete | Production | QFN (RTJ) \| 20 | - | - | Call TI | Call TI | -40 to 125 | -----> DRV425 |

(1) **Status:** For more details on status, see our product life cycle.

(2) **Material type:** When designated, preproduction parts are prototypes/experimental devices, and are not yet approved or released for full production. Testing and final process, including without limitation quality assurance, reliability performance testing, and/or process qualification, may not yet be complete, and this item is subject to further changes or possible discontinuation. If available for ordering, purchases will be subject to an additional waiver at checkout, and are intended for early internal evaluation purposes only. These items are sold without warranties of any kind.

(3) **RoHS values:** Yes, No, RoHS Exempt. See the TI RoHS Statement for additional information and value definition.

(4) **Lead finish/Ball material:** Parts may have multiple material finish options. Finish options are separated by a vertical ruled line. Lead finish/Ball material values may wrap to two lines if the finish value exceeds the maximum column width.

(5) **MSL rating/Peak reflow:** The moisture sensitivity level ratings and peak solder (reflow) temperatures. In the event that a part has multiple moisture sensitivity ratings, only the lowest level per JEDEC standards is shown. Refer to the shipping label for the actual reflow temperature that will be used to mount the part to the printed circuit board.

(6) **Part marking:** There may be an additional marking, which relates to the logo, the lot trace code information, or the environmental category of the part.

Multiple part markings will be inside parentheses. Only one part marking contained in parentheses and separated by a "~" will appear on a part. If a line is indented then it is a continuation of the previous line and the two combined represent the entire part marking for that device.

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In no event shall TI's liability arising out of such information exceed the total purchase price of the TI part(s) at issue in this document sold by TI to Customer on an annual basis.

**OTHER QUALIFIED VERSIONS OF DRV425 :**

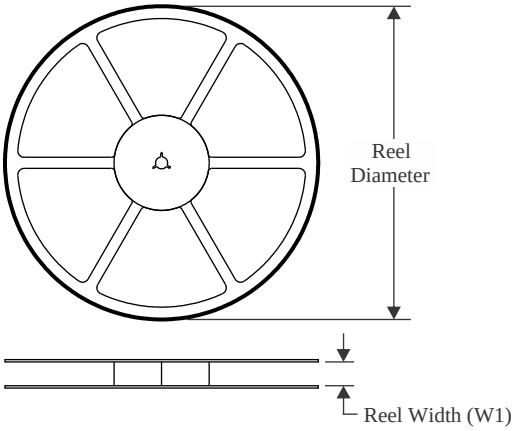
- Automotive : DRV425-Q1

NOTE: Qualified Version Definitions:

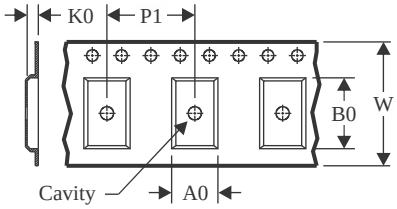
- Automotive - Q100 devices qualified for high-reliability automotive applications targeting zero defects

## TAPE AND REEL INFORMATION

**REEL DIMENSIONS**

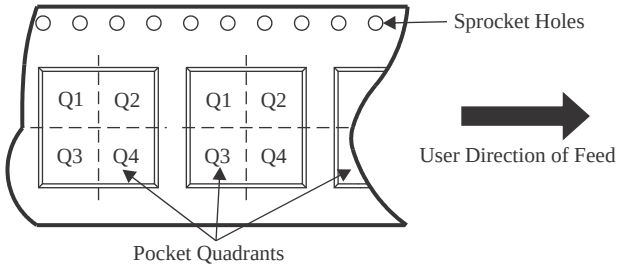



**TAPE DIMENSIONS**




| | |
|---|---|
| A0 | Dimension designed to accommodate the component width |
| B0 | Dimension designed to accommodate the component length |
| K0 | Dimension designed to accommodate the component thickness |
| W | Overall width of the carrier tape |
| P1 | Pitch between successive cavity centers |

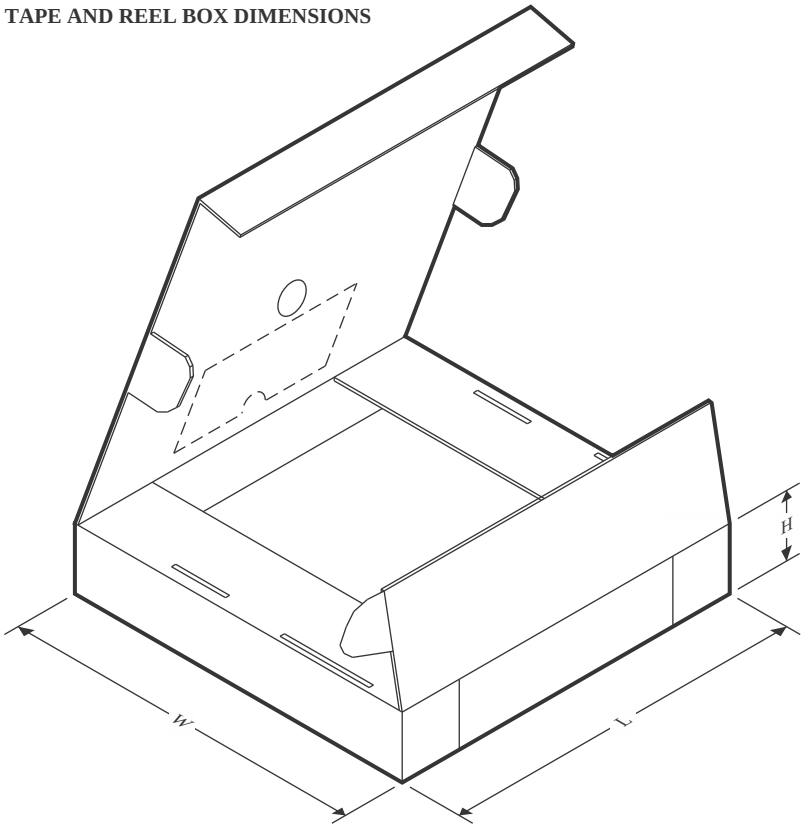
**QUADRANT ASSIGNMENTS FOR PIN 1 ORIENTATION IN TAPE**




*All dimensions are nominal

| Device | Package Type | Package Drawing | Pins | SPQ | Reel Diameter (mm) | Reel Width W1 (mm) | A0 (mm) | B0 (mm) | K0 (mm) | P1 (mm) | W (mm) | Pin1 Quadrant |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| DRV425RTJR | QFN | RTJ | 20 | 3000 | 330.0 | 12.4 | 4.25 | 4.25 | 1.15 | 8.0 | 12.0 | Q2 |

TAPE AND REEL BOX DIMENSIONS




*All dimensions are nominal

| Device | Package Type | Package Drawing | Pins | SPQ | Length (mm) | Width (mm) | Height (mm) |
|---|---|---|---|---|---|---|---|
| DRV425RTJR | QFN | RTJ | 20 | 3000 | 346.0 | 346.0 | 33.0 |

# GENERIC PACKAGE VIEW

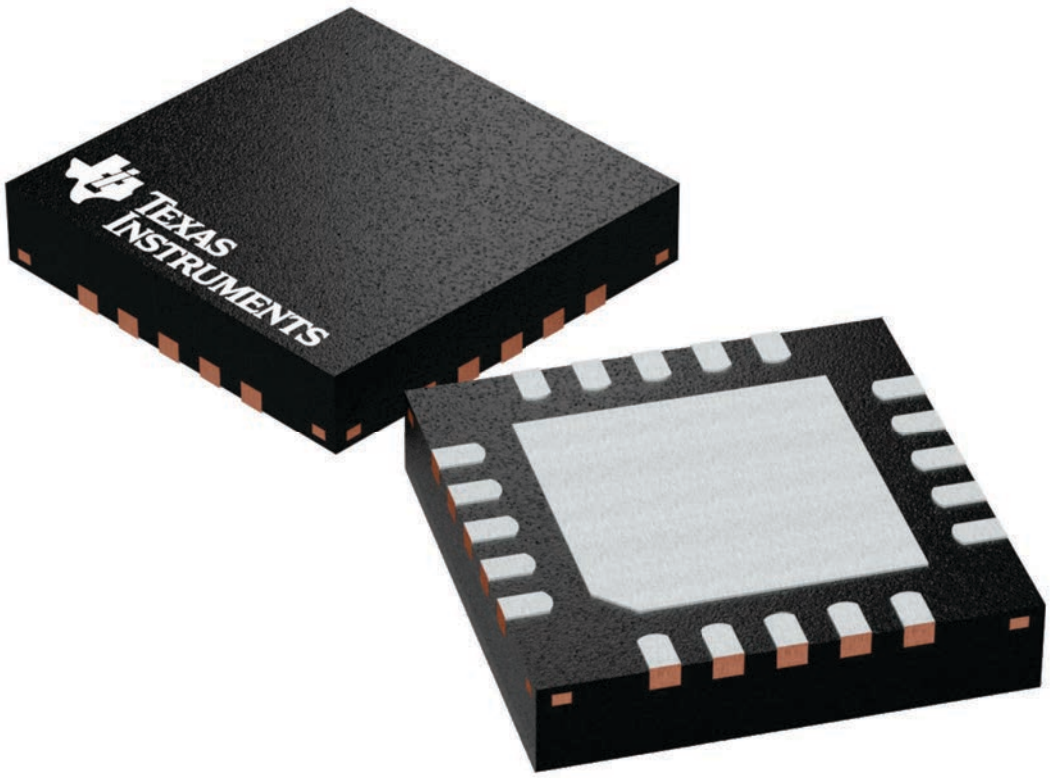
## RTJ 20

## WQFN - 0.8 mm max height

**4 x 4, 0.5 mm pitch**

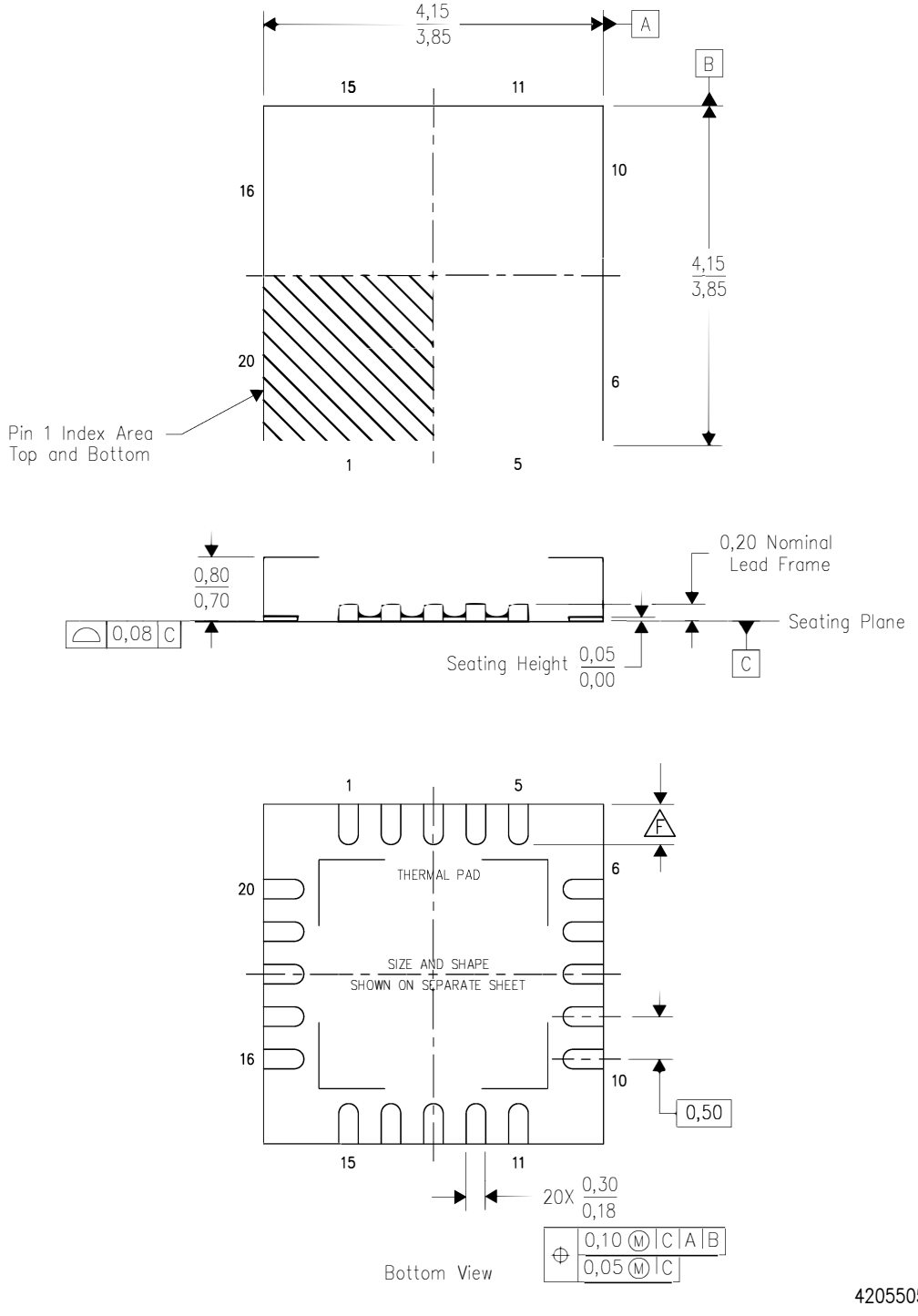
PLASTIC QUAD FLATPACK - NO LEAD

This image is a representation of the package family, actual package may vary.
Refer to the product data sheet for package details.




4224842/A

RTJ (S–PWQFN–N20) PLASTIC QUAD FLATPACK NO–LEAD




NOTES:
A. All linear dimensions are in millimeters. Dimensioning and tolerancing per ASME Y14.5–1994.
B. This drawing is subject to change without notice.
C. QFN (Quad Flatpack No–Lead) package configuration.
D. The package thermal pad must be soldered to the board for thermal and mechanical performance.
E. See the additional figure in the Product Data Sheet for details regarding the exposed thermal pad features and dimensions.
F. Check thermal pad mechanical drawing in the product datasheet for nominal lead length dimensions.

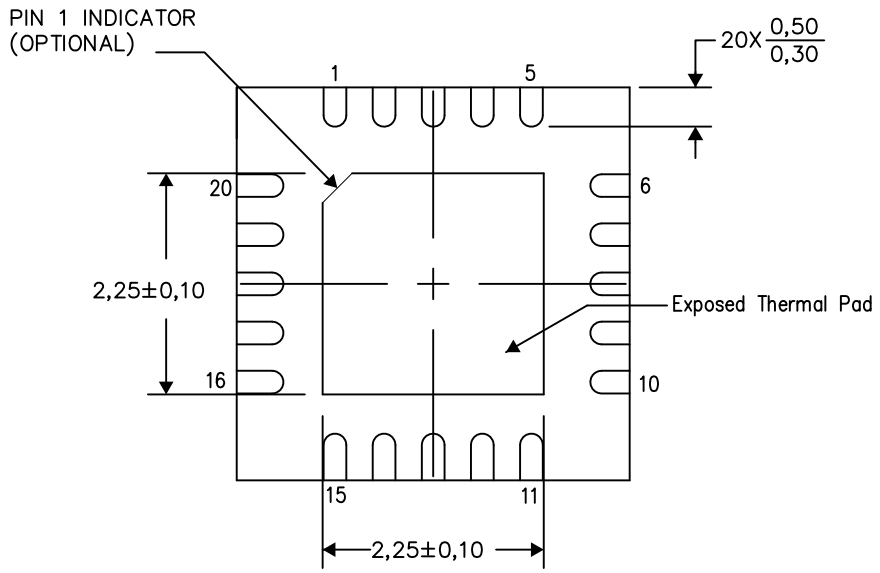
RTJ (S–PWQFN–N20) PLASTIC QUAD FLATPACK NO–LEAD

THERMAL INFORMATION

This package incorporates an exposed thermal pad that is designed to be attached directly to an external heatsink. The thermal pad must be soldered directly to the printed circuit board (PCB). After soldering, the PCB can be used as a heatsink. In addition, through the use of thermal vias, the thermal pad can be attached directly to the appropriate copper plane shown in the electrical schematic for the device, or alternatively, can be attached to a special heatsink structure designed into the PCB. This design optimizes the heat transfer from the integrated circuit (IC).

For information on the Quad Flatpack No–Lead (QFN) package and its advantages, refer to Application Report, QFN/SON PCB Attachment, Texas Instruments Literature No. SLUA271. This document is available at www.ti.com.

The exposed thermal pad dimensions for this package are shown in the following illustration.




Bottom View

Exposed Thermal Pad Dimensions

4206256–8/V 05/15

NOTE: All linear dimensions are in millimeters

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